

Heterogeneous Households and Aggregate Dynamics: Belief-Invariant Analytical Moment Recursion[†]

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Abstract

We develop a heterogeneous-household framework with closed-form aggregate dynamics that jointly capture growth, business cycles, and distributional change. Three moments—aggregate capital and two wealth-sorting covariances—are sufficient for the equilibrium dynamics, and the equilibrium is invariant to household beliefs. The closed-form structure enables exact inversion of U.S. data on output, capital, and hours (1950–2019) into structural shocks in general equilibrium. The inversion reveals a *sorting accelerator*: TFP growth compounds near-unit-root return-wealth sorting, and the resulting distributional effect grows with shock persistence. This channel is decisive in counterfactual capital-tax reforms. The recovered shocks also trace a two-dimensional business cycle, in which TFP is nearly orthogonal to a recurrent labor-efficiency and patience pairing.

Keywords: Heterogeneous agents, closed-form aggregation, wealth inequality, business cycles, belief invariance.

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1 Introduction

Wealth inequality in the United States has risen steadily since the 1950s, yet the economy has also experienced recurring business cycles and sustained long-run growth. These three phenomena are deeply intertwined. TFP growth raises returns disproportionately for wealthy households with favorable portfolio draws, amplifying inequality through persistent return-wealth sorting. The resulting concentration, in turn, makes the macroeconomy more sensitive to aggregate shocks. Understanding this feedback requires a framework that handles growth, cycles, and distributional dynamics jointly along a non-stationary path.

This paper constructs such a framework. We show that in a heterogeneous-household economy with idiosyncratic productivity and return risk, three summary statistics—aggregate capital and two covariances measuring the alignment of wealth with productivity and portfolio returns—are sufficient for all contemporaneous equilibrium allocations and prices, and aggregate capital follows an exact closed-form law of motion. We call the resulting object a *moment-recursive competitive equilibrium* (MRCE).

The paper makes three contributions. First, we derive closed-form policies for consumption, saving, and labor supply under incomplete markets with idiosyncratic productivity and return risk. These yield a unique MRCE where $\mathcal{M} = [K, \text{cov}(z, a), \text{cov}(e, a)]$ is sufficient for all contemporaneous equilibrium allocations and prices, and the aggregate capital law of motion K' is an exact closed-form function of $\mathcal{M}$. The saving rate is increasing and concave in wealth (Dynan et al., 2004). The framework nests several benchmarks: shutting down return heterogeneity yields Krusell–Smith; shutting down all heterogeneity recovers Brock and Mirman (1972). With return risk alone, the wealth distribution drops out as in Angeletos (2007); adding productivity heterogeneity restores its role through exactly two covariances.

Second, the equilibrium is invariant to household beliefs about aggregate dynamics. Under log utility, income and substitution effects cancel exactly, so policies depend only on current prices. This eliminates the fixed-point problem over beliefs, making the MRCE a belief-invariant recursive competitive equilibrium (BRCE). The BRCE accommodates non-stationary paths without detrending—essential for studying how TFP *growth*, not just TFP levels, drives inequality. Conventional approaches either compare stationary equilibria at

different TFP levels or require expectations over the entire future price sequence. The BRCE sidesteps both: any TFP path can be fed into the forward simulation to obtain valid equilibrium dynamics.

Third, the closed-form structure enables an *exact general equilibrium inversion* that operates at two scales. At the aggregate scale, given U.S. data on output, capital, and hours (1950–2019), we recover TFP, time-varying patience, and labor productivity that exactly rationalize the data period by period—extending the business-cycle accounting of Chari et al. (2007) to a heterogeneous-agent setting with structural rather than reduced-form shocks. At the household scale, taking the recovered aggregate prices as given, the same closed-form policies invert PSID and SCF micro data into individual structural parameters (z_i, e_i, β_i) wave by wave. The two scales nest: the macro β_t recovered from aggregates equals (up to sample selection) the capital-weighted aggregate of household β_i recovered from surveys, and PSID and SCF—two independent surveys with disjoint samples and orthogonal identification strategies—agree on this value within 1.7 pp. Because the recovered objects are structural parameters in a model with explicit distributional dynamics, they support counterfactual decompositions that isolate each shock’s role in inequality growth, deliver conditional forecasts that outperform a recursive AR benchmark for output and labor, and pass a Lucas-critique-style test that no household’s β_i shifts around four major federal tax reforms.

The inversion yields several substantive findings. A *sorting accelerator* channel links TFP growth to wealth concentration: higher TFP raises returns, which compound through near-unit-root return-wealth sorting and widen the wealth distribution via self-reinforcing feedback (Piketty, 2014). At the baseline calibration, the channel’s unconditional amplification of a one-period TFP shock is modest. It becomes quantitatively decisive under persistent policy shifts: the 1981 and 2001 capital-tax reforms produce wealth-distribution divergences that widen for decades, because the near-unit-root sorting mechanism prevents mean reversion (Section 4.6). The recovered shocks also trace a two-dimensional business cycle. TFP is nearly orthogonal to both labor productivity and patience ($\rho(\Delta \log A, \Delta \log Z_L) = 0.142$, $\rho(\Delta \log A, \Delta \beta) = 0.105$). Labor productivity and patience, by contrast, are strongly negatively correlated ($\rho(\Delta \log Z_L, \Delta \beta) = -0.858$): labor-efficiency declines coincide with rising patience, an anti-cyclical pairing consistent with a precautionary-saving channel. The struc-

tural classification attributes individual postwar recessions to different shocks, with this labor-efficiency/patience pairing appearing as one recognizable but not universal pattern.

Related literature [Bewley \(1986\)](#), [Huggett \(1993\)](#), and [Aiyagari \(1994\)](#) developed the foundational heterogeneous-agent framework. [Krusell and Smith \(1998\)](#) showed numerically that mean wealth nearly suffices for aggregate forecasting; the MRCE identifies the omitted moments and gives exact conditions under which the approximation succeeds. The result is closely related to [Den Haan and Rendahl \(2010\)](#), who show that cross-sectional averages of individual-policy basis functions are natural aggregate states. Here those averages reduce to economically interpretable sorting covariances. Computational approaches to high-dimensional heterogeneous-agent dynamics include [Reiter \(2009\)](#), [Winberry \(2018\)](#), [Ahn et al. \(2018\)](#), and the sequence-space Jacobian method of [Auclert et al. \(2021\)](#). The contribution here is complementary: it gives exact nonlinear dynamics for a restricted class rather than linearized dynamics for a broad class.

The paper also builds on exact aggregation under log preferences. [Brock and Mirman \(1972\)](#) solve the representative-agent growth model, and [Angeletos \(2007\)](#) obtains aggregation with uninsured return risk. This paper shows how adding labor-productivity heterogeneity restores a role for the wealth distribution through two covariances. [Bilbiie \(2020\)](#) and [Ravn and Sterk \(2021\)](#) develop parsimonious heterogeneous-agent models with analytical aggregate dynamics; the MRCE uses a continuum of agents with rich return heterogeneity. [Alvarez and Lippi \(2022\)](#) develop analytical impulse responses using eigenvalue methods in menu-cost models; the moment structure here plays an analogous role for heterogeneous households.

[Krueger et al. \(2025\)](#) and [Ando et al. \(2025\)](#) achieve closed-form transition paths in a limited-commitment economy where the zero-income state reduces each type’s problem to cake-eating under log utility. The MRCE takes a different tractability route from the same log-utility foundation: rather than eliminating labor income, it folds labor income into the effective return through a Stone–Geary wealth-leisure complementarity, preserving continuous idiosyncratic distributions, return heterogeneity, and endogenous labor supply. The trade-off is that the limited-commitment structure delivers a closed-form transition for

K alone, while the MRCE’s covariance dynamics require agent-based simulation. The two approaches are complementary: they are dual entry points into analytical heterogeneous-agent macro, each covering ground the other cannot.

The belief-invariance result is related to block-recursive equilibria (Menzio and Shi, 2010) and cognitive discounting (Gabaix, 2020), but arises endogenously from the income-substitution offset under log utility—the same property that Krueger et al. (2025) and Ando et al. (2025) exploit when noting that current policies are independent of future interest rates. Belief invariance is thus a shared engine across multiple tractable heterogeneous-agent structures, not a feature unique to any one market arrangement. The state-dependent amplification results contribute to the nonlinear business cycle literature: Storesletten et al. (2004) document countercyclical cross-sectional risk, Bloom et al. (2018) study uncertainty-driven state dependence, and Berger and Vavra (2015) analyze wealth heterogeneity in consumption dynamics. The asset pricing results relate to idiosyncratic risk and equilibrium returns (Constantinides and Duffie, 1996; Schmidt, 2016); growing evidence on heterogeneous portfolio returns (Fagereng et al., 2020; Bach et al., 2020) motivates the model’s return channel.

Roadmap The remainder of the paper proceeds as follows. Section 2 sets up the model. Section 3 derives the MRCE, establishes belief invariance, and draws out its macroeconomic and asset pricing implications. Section 4 implements the exact inversion on U.S. aggregate data and evaluates the model’s quantitative predictions. Section 5 extends the inversion to the household level, recovering individual structural parameters from PSID and SCF and validating the macro β_t as a capital-weighted aggregate of micro β_i . Section 6 concludes. Proofs, extensions, and additional applications are collected in the Online Appendix.

2 Model

A continuum of infinitely lived households face idiosyncratic labor productivity and portfolio return risk and choose consumption, saving, and labor supply each period.

2.1 States and shocks

Each household's state is $x = [a, z, e]$: wealth a (endogenous), idiosyncratic labor productivity z (exogenous), and a return-scaling factor e (exogenous). All three display substantial cross-sectional variation (Kuhn et al., 2020; Storesletten et al., 2004; Fagereng et al., 2020).

The idiosyncratic processes follow stationary AR(1) dynamics in levels with unit mean:

$$z' - 1 = \rho_z (z - 1) + \varepsilon'_z, \quad \varepsilon'_z \sim \mathcal{N}(0, \sigma_z^2), \quad (1)$$

$$e' - 1 = \rho_e (e - 1) + \varepsilon'_e, \quad \varepsilon'_e \sim \mathcal{N}(0, \sigma_e^2), \quad (2)$$

where σ_z, σ_e are the per-period *innovation* standard deviations (matching the calibration code), and $\varepsilon'_z, \varepsilon'_e$ are mutually independent. The unconditional variances are $\sigma_z^2/(1 - \rho_z^2)$ and $\sigma_e^2/(1 - \rho_e^2)$. Linearity in levels is essential: it ensures that the cross-sectional covariances $\text{cov}(z, a)$ and $\text{cov}(e, a)$ serve as sufficient statistics for contemporaneous equilibrium allocations (Section 3). In the quantitative analysis, each process is discretized using the Tauchen (1986) method.

Each household's effective labor productivity per hour is $Z_L z$, where Z_L is an aggregate labor productivity parameter and z is the idiosyncratic component ($\mathbb{E}[z] = 1$). Z_L captures aggregate labor market conditions: utilization intensity, matching quality, and workforce composition. It enters the production function only through the efficiency-weighted labor aggregate $L \equiv Z_L \int z n(x; X) d\Phi$, introduced in Section 2 below; it does not appear as a separate factor multiplying $L^{1-\alpha}$. This asymmetry distinguishes Z_L from neutral TFP A : a shift in Z_L rescales effective hours one-for-one but does not change the production technology itself (see Section 4 and Online Appendix B for the structural interpretation). At steady state, $Z_L = 1$.

The aggregate state is $X = [\Phi, A, Z_L, \beta, B, C]$: the cross-sectional distribution Φ , aggregate TFP A , aggregate labor productivity Z_L , the discount factor β , return dispersion B , and productivity dispersion C . In the quantitative application (Section 4), A , Z_L , and β vary; B and C are held at steady state. The framework accommodates time-variation in all five—e.g., widening return dispersion during financial crises or shifting patience during demand-driven episodes.

2.2 Preferences

Household preferences take a Stone-Geary form with endogenous labor supply:

$$u(c, n, a, z; X) = \log(c - g(x; X)) + \theta \log((\kappa + \xi a) - n), \quad (3)$$

where c is consumption, n is labor supply, and $\theta > 0$ governs the relative weight on leisure.

The subsistence level $g(x; X) = \kappa w(X)(1 - \tau)Z_L z$ captures work-related expenses that scale with after-tax earnings.

The effective time endowment $\kappa + \xi a$ increases in wealth: $\kappa \in (0, 1)$ is the base endowment, and $\xi > 0$ captures wealth buying usable time through time-saving services ([Aguiar and Hurst, 2007](#)). This wealth-leisure complementarity links the wealth distribution to aggregate labor supply.

The parameters (θ, κ, ξ) are calibrated to match average hours, the saving-to-labor-income ratio, and the level of wealth concentration.

2.3 Budget constraint and the household problem

The recursive formulation of the household's problem is:

$$v(x; X) = \max_{c, a', n} u(c, n, a, z; X) + \beta \mathbb{E}v(x'; X') \quad (4)$$

$$\text{s.t.} \quad c + a' = w(X)(1 - \tau)Z_L z n + (1 + r(X)(1 - \tau)e)a \quad (5)$$

$$\Phi' = \Gamma_\Phi(X) \quad (6)$$

$$a' \geq 0, \quad n \geq 0 \quad (7)$$

The household's problem does not include the transfer T : households are unaware of its existence and optimize based solely on factor income. The actual law of motion for wealth is $a'_{\text{actual}} = a'_{\text{chosen}} + T(X)$, where T is added by the government after the household's decision (see Section 2.4). The wage $w(X)$ is per hour of labor and $r(X)$ is the aggregate return on capital. The factor e scales returns: $\mathbb{E}[e] = 1$, so $r(X)$ is the economy-wide average. The product $Z_L z$ determines the household's effective hourly wage; Z_L captures aggregate labor

market conditions while z captures individual heterogeneity. The discount factor $\beta \in (0, 1)$ is an element of the aggregate state X and may vary over time, capturing shifts in patience due to financial conditions, uncertainty, or other demand-side forces. The household faces a borrowing constraint $a' \geq 0$, which is non-binding in equilibrium (shown below).

2.4 Production

A representative firm operates a constant-returns-to-scale production technology:

$$Y = AK_{eff}^\alpha L^{1-\alpha}, \quad (8)$$

where the inputs are *efficiency-weighted* factor aggregates:

$$K_{eff} \equiv \int e a d\Phi = K + \text{cov}(e, a), \quad L \equiv Z_L \int z n(x; X) d\Phi, \quad (9)$$

with $K = \int a d\Phi$ denoting physical (wealth) capital and $\alpha \in (0, 1)$. The return factor e_i scales the efficiency of household i 's capital in production (e.g., reflecting portfolio quality or entrepreneurial ability), just as z_i scales labor productivity. Factor prices satisfy:

$$r(X) = \alpha AK_{eff}^{\alpha-1} L^{1-\alpha} - \delta, \quad w(X) = (1 - \alpha) AK_{eff}^\alpha L^{-\alpha}, \quad (10)$$

where δ is the depreciation rate. Competitive factor markets pay $(r + \delta)$ per efficiency unit of capital and w per efficiency unit of labor. Household i supplies $e_i a_i$ efficiency units of capital and $Z_L z_i n_i$ efficiency units of labor, earning $(r + \delta)e_i a_i - \delta a_i$ and $w Z_L z_i n_i$ respectively. Aggregating: total capital income $(r + \delta)K_{eff} = \alpha Y$ and total labor income $wL = (1 - \alpha)Y$, exhausting output. Because factor prices depend on K_{eff}/L , distributional forces—both labor supply shifts and return-wealth sorting—feed back into prices.

2.5 Government policy

The government levies proportional income taxes—at rate τ_K on capital income and τ_L on labor income—and redistributes a fraction η of tax revenue as a lump-sum transfer to all

households:

$$T(X) = \eta \cdot [\tau_K(r + \delta)K_{eff} + \tau_L wL], \quad (11)$$

When $\tau_K = \tau_L = \tau$, this reduces to $T = \eta\tau Y$ (since $(r + \delta)K_{eff} + wL = Y$). After-tax prices are $\tilde{r} = r(1 - \tau_K)$ and $\tilde{w} = w(1 - \tau_L)$.¹ Remaining revenue finances government consumption (not in utility or production). The baseline sets $\tau_K = \tau_L = 0.30$; the Online Appendix considers time-varying rates.

The transfer plays a dual role. First, it is a fiscal policy channel: tax revenue is partially redistributed, with η controlling the degree of redistribution. Higher η compresses the wealth distribution and thins the Pareto tail. Second, it provides the additive intercept in the Kesten process governing wealth dynamics, which is necessary for a stationary distribution with a Pareto tail to exist (Kesten, 1973). Without $T > 0$, the purely multiplicative system $a' = \beta R_i a$ collapses to the degenerate distribution at zero and admits no non-degenerate stationary distribution.

Timing The transfer arrives after the household’s consumption-saving decision: households solve their problem without knowledge of T , and the government adds T to wealth at the end of the period. This timing is essential for preserving the analytical tractability. If households anticipated T , they would incorporate its present value into effective wealth, breaking the $v = A \log a + B$ value function conjecture and the multiplicative-additive separation $a' = \beta R_i a + T$ that enables closed-form aggregation. The assumption is quantitatively innocuous: at the baseline calibration, the per-period transfer is 0.1% of mean wealth, and the present value of the entire transfer stream is approximately 2.6% of mean wealth. Incorporating this present value into effective wealth would shift saving shares by a correspondingly small amount.

¹Government revenue in formula (11) is computed on gross capital income $(r + \delta)K_{eff}$ — consistent with U.S. corporate taxation of book profits net of depreciation — while the household’s after-tax return $\tilde{r} = r(1 - \tau_K)$ is defined on the net return r . The implied wedge between household tax payments and aggregate revenue is $\tau_K \delta K_{eff}$, equal to about 1.5 pp of capital income at baseline ($\tau_K = 0.30$, $\delta = 0.07$). This wedge is absorbed in the lump-sum transfer T and does not affect any household first-order condition or aggregation result; we retain the formulation because it preserves the linear separability $a' = \beta R_i a + T$ on which Theorem 1 depends.

2.6 Recursive competitive equilibrium

Definition 1 (Recursive competitive equilibrium (RCE)).

A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, w, r)$ is a recursive competitive equilibrium if:

1. g_c, g_a, g_n, v solve the household's problem given Γ_Φ .
2. K, L satisfy the production sector's optimality conditions.
3. Factor markets clear.
4. Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the intertemporal policy function g_a and the exogenous idiosyncratic state processes to Φ .

The difficulty is that Φ is infinite-dimensional. The next section shows that optimal policies depend on Φ only through three moments—aggregate capital and two covariances—that serve as sufficient statistics for all contemporaneous equilibrium allocations.

3 Moment-Recursive Competitive Equilibrium

3.1 Analytical household policies

The household's optimal policies have explicit solutions that reduce the infinite-dimensional distribution to a finite moment vector.

Proposition 1 (Analytical household policies and wealth dynamics).

The optimal consumption and labor supply policies are:

$$c(x; X) = \frac{1 - \beta}{1 + \theta} (1 + r(X)(1 - \tau)e)a + \left(\frac{1 - \beta}{1 + \theta} a\xi + \kappa \right) w(X)(1 - \tau)Z_L z \quad (12)$$

$$n(x; X) = \max \left\{ \kappa(1 + \theta) + \xi a - \frac{\theta c(x; X)}{w(X)(1 - \tau)Z_L z}, 0 \right\} \quad (13)$$

The household's chosen saving is $\hat{a}'(x; X) = \beta a ((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z\xi)$.

The actual law of motion for wealth, after the unanticipated government transfer, is:

$$a'(x; X) = \hat{a}'(x; X) + T(X) = \beta a((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z \xi) + T(X). \quad (14)$$

Proof. The budget constraint can be rewritten as:

$$(1 + \theta)(c - \kappa Z_L z w(X)(1 - \tau)) + a' = ((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z \xi) a. \quad (15)$$

Under log utility, the standard result of [Brock and Mirman \(1972\)](#) yields $a' = \beta a R_i$. The government then adds the unanticipated transfer: $a'_{\text{actual}} = \beta a R_i + T(X)$. ■

Consumption is linear in wealth with source-dependent MPCs. The saving function $a' = \beta a(\cdot) + T(X)$ separates into a multiplicative component (enabling aggregation) and an additive transfer (ensuring stationarity).

3.2 The saving rate

The saving rate has a closed-form characterization.

Proposition 2 (Analytical saving rate).

The saving rate $s(x; X) = \frac{a'(x; X) - a}{y(x; X)}$, where $y(x; X) = w(X)(1 - \tau)Z_L z n(x; X) + r(X)(1 - \tau)ea + T(X)$ is after-tax income plus transfer, satisfies:

$$s(x; X) = \frac{a[\beta(1 + \tilde{r}e + \tilde{w}Z_L z \xi) - 1] + T}{\tilde{w}Z_L z \kappa \theta + a \left[\frac{\tilde{w}Z_L z \xi(1 + \theta \beta)}{1 + \theta} - \frac{\theta(1 - \beta)(1 + \tilde{r}e)}{1 + \theta} + \tilde{r}e \right] + T}, \quad (16)$$

where $\tilde{r} = r(X)(1 - \tau)$ and $\tilde{w} = w(X)(1 - \tau)$, with the following properties:

- (i) $\lim_{a \rightarrow \infty} s(x; X) = \frac{\beta(1 + \tilde{r}e + \tilde{w}Z_L z \xi) - 1}{\frac{\tilde{w}Z_L z \xi(1 + \theta \beta)}{1 + \theta} - \frac{\theta(1 - \beta)(1 + \tilde{r}e)}{1 + \theta} + \tilde{r}e}$
- (ii) $\lim_{a \rightarrow 0} s(x; X) > 0$ due to the transfer $T(X) > 0$
- (iii) $\frac{\partial s}{\partial a} > 0$ and $\frac{\partial^2 s}{\partial a^2} < 0$.

Proof. See Appendix A.2 of the Online Appendix. ■

The saving rate is positive at zero wealth (due to T), increasing, and concave—the pattern documented by [Dynan et al. \(2004\)](#). Concavity generates mean reversion in relative wealth: as wealth rises, the subsistence floor absorbs a diminishing share and the saving rate converges to a constant.

Non-binding borrowing constraint An important consequence of the policy functions is that the borrowing constraint is always slack:

Corollary 1 (Non-binding borrowing constraint).

Along the realized wealth path, the actual law of motion satisfies $a'(x; X) \geq T(X) > 0$, so the borrowing limit $a' \geq 0$ is non-binding for all households at every date.

Proof. The household's chosen saving $\hat{a}'(x; X) = \beta a((1+r(X)(1-\tau)e)+w(X)(1-\tau)Z_L z \xi) \geq 0$ for all $a \geq 0$, with equality only on the measure-zero set $\{a = 0\}$. The actual law of motion adds the unanticipated transfer: $a'(x; X) = \hat{a}'(x; X) + T(X) \geq T(X) > 0$. Hence the constraint is strictly slack along the realized wealth distribution. ■

Non-binding constraints keep policies globally linear, enabling analytical aggregation. This is a scope condition: the MRCE applies when transfers are large enough relative to borrowing needs that the constraint never binds. Where binding constraints create kinks, the analytical aggregation breaks down. The condition is broadly consistent with the U.S. economy given Social Security, Medicare, and other social insurance.

3.3 Existence and characterization

Definition 2 (Moment-recursive competitive equilibrium (MRCE)).

A collection $(g_c, g_a, g_n, v, \Gamma_M, K, L, w, r, \mathcal{M})$ is a moment-recursive competitive equilibrium if:

1. g_c, g_a, g_n, v solve the household's problem given Γ_M .
2. K, L satisfy the production sector's optimality conditions.
3. Factor markets clear.

4. Γ_M satisfies $\Gamma_M(\mathcal{M}(\Phi), A, Z_L, \beta) = \mathcal{M}(\Phi')$, where $\mathcal{M}$ extracts the sufficient moments from Φ , and Φ' is the distribution of individual states induced by g_a and the exogenous processes.

The key departure from the RCE is that Γ_M maps a *finite* moment vector to contemporaneous equilibrium allocations and prices. The sufficient statistics are aggregate capital K and two covariances: $\text{cov}(z, a)$ measures the alignment between earnings ability and wealth; $\text{cov}(e, a)$ measures the alignment between portfolio returns and wealth.

Proposition 3 (Unique existence of the MRCE).

The MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, \text{cov}(z, a), \text{cov}(e, a)]$. Define after-tax prices $\tilde{r} = r(\mathcal{M}, A)(1 - \tau)$, $\tilde{w} = w(\mathcal{M}, A)(1 - \tau)$, and the transfer $T = \eta \cdot Y \cdot \tau$. Then:

(i) *Contemporaneous sufficiency.* Given $\mathcal{M}$ and the aggregate state (A, Z_L, β) , all equilibrium allocations—individual consumption $c(x; X)$, saving $a'(x; X)$, labor supply $n(x; X)$, and aggregate prices r, w, L —are fully determined without reference to the distribution Φ beyond its moments $\mathcal{M}$.

(ii) *Aggregate capital law of motion.* K' has an exact closed-form expression:

$$K' = \beta [(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi \text{cov}(z, a) + \tilde{r} \text{cov}(e, a)] + T \quad (17)$$

(iii) *Factor prices and aggregate labor.* Defining $K_{eff} = K + \text{cov}(e, a)$:

$$r(\mathcal{M}, A) = \alpha A \left(\frac{L}{K_{eff}} \right)^{1-\alpha} - \delta \quad (18)$$

$$w(\mathcal{M}, A) = (1 - \alpha) A \left(\frac{K_{eff}}{L} \right)^\alpha \quad (19)$$

$$L(\mathcal{M}, A, Z_L, \beta) = Z_L \kappa + \frac{Z_L \xi (1 + \theta \beta) (K + \text{cov}(z, a))}{1 + \theta} - \frac{\theta(1 - \beta)}{(1 + \theta)\tilde{w}} [K(1 + \tilde{r}) + \text{cov}(e, a)\tilde{r}] \quad (20)$$

Proof. See Appendix A.3 of the Online Appendix. ■

The K' equation reveals three distinct channels. The first term $(1 + \tilde{r} + \tilde{w}Z_L\xi)K$ is the representative-agent channel. The second, $\tilde{r} \text{cov}(e, a)$, is the *return-sorting channel*: high-

return households are disproportionately wealthy, raising aggregate saving above what average returns and average wealth would predict. The third, $\tilde{w}Z_L\xi \text{cov}(z, a)$, is the *productivity-sorting channel*: productive-wealthy households generate additional saving through wealth-leisure complementarity. These terms make dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

Distributional dynamics and the closure problem While $\mathcal{M}$ is sufficient for all contemporaneous equilibrium allocations and K' is exactly determined, the forward evolution of $\text{cov}(z', a')$ and $\text{cov}(e', a')$ generally depends on higher-order moments of the joint distribution Φ . Specifically, from $a' = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L z\xi) + T$ and $z' = \rho_z z + \varepsilon_z$:

$$\text{cov}(z', a') = \rho_z \beta [(1 + \tilde{r} + \tilde{w}Z_L\xi) \text{cov}(z, a) + \tilde{w}Z_L\xi \mathbb{E}[\hat{z}^2 a] + \tilde{r} \mathbb{E}[\hat{z}\hat{e}a]], \quad (21)$$

where $\hat{z} = z - 1$ and $\hat{e} = e - 1$. The terms $\mathbb{E}[\hat{z}^2 a]$ and $\mathbb{E}[\hat{z}\hat{e}a]$ involve three-way cross-moments that are not functions of $\mathcal{M}$ alone. Under joint normality of $(\hat{z}, \hat{e}, a)$, these reduce to $\mathbb{E}[\hat{z}^2 a] = \sigma_z^2 K$ and $\mathbb{E}[\hat{z}\hat{e}a] = 0$ (by the Isserlis/Wick theorem, using independence of z and e), and the moment dynamics close. However, the Kesten-process wealth dynamics generate Pareto tails that violate normality.

Therefore, tracking the full distribution via agent-based simulation is necessary for the forward evolution of distributional moments, while the exact household policies, the K' equation (17), and all contemporaneous allocations are preserved analytically. The quantitative analysis in Section 4 employs this approach: exact policies govern each agent's behavior, and the distributional moments are computed from the simulated cross-section.

3.4 Ergodicity

Corollary 2 (Kesten stationarity).

The individual law of motion $a' = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L z\xi) + T$ is a Kesten process with multiplicative coefficient $R_i = \beta((1 + \tilde{r}e) + \tilde{w}Z_L z\xi)$ and additive intercept $T > 0$. When $\mathbb{E}[\log R_i] < 0$, the wealth distribution admits a unique stationary solution with Pareto tails (Kesten, 1973). The additive transfer $T > 0$ is essential: without it, the purely multiplicative

system admits only the degenerate distribution at zero.

Proof. See Appendix A.4 of the Online Appendix. ■

The near-unit-root property of the dominant eigenvalue (empirically ≈ 1.01 in the calibrated model) generates slow mean reversion of the wealth distribution and amplifies shock propagation, as shown in Section 4.

3.5 Special cases

The MRCE nests several benchmark models, clarifying what each source of heterogeneity contributes to aggregate dynamics.

Corollary 3 (Krusell and Smith (1998) analogue).

Suppose return heterogeneity is muted: $e = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, \text{cov}(z, a)]$, and the aggregate capital law of motion is:

$$K' = \beta [(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi \text{cov}(z, a)] + T \quad (22)$$

Proof. See Appendix A.5 of the Online Appendix. ■

Without return heterogeneity, the state reduces to $[K, \text{cov}(z, a)]$ —the analytical counterpart of Krusell and Smith (1998). Capital alone suffices if and only if $\text{cov}(z, a) = 0$. Approximate aggregation holds when $\text{cov}(z, a)$ is small relative to K —i.e., when productivity shocks are weakly persistent or wealth mobility is high. The MRCE thus provides exact conditions under which the Krusell and Smith (1998) finding holds as a theorem. Further discussion appears in Appendix B.3.

Corollary 4 (Representative-agent analogue).

Suppose all heterogeneity is muted: $e = 1$ and $z = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = K$, and the law of motion is:

$$K' = \beta [(1 + \tilde{r} + \tilde{w}Z_L\xi)K] + T, \quad (23)$$

which leads to the aggregate consumption function:

$$C(\mathcal{M}, A, Z_L, \beta) = \frac{1 - \beta}{1 + \theta} (1 + \tilde{r} + \tilde{w} Z_L \xi) K + \kappa \tilde{w} Z_L. \quad (24)$$

Proof. See Appendix A.5 of the Online Appendix. ■

With all heterogeneity eliminated, both covariances vanish and the model reduces to the stochastic growth model with log preferences and endogenous labor supply (Brock and Mirman, 1972). Every departure of the full MRCE from this benchmark is therefore attributable to sorting between wealth and idiosyncratic characteristics.

3.6 Belief invariance

A distinctive feature of the MRCE is invariance to household beliefs about aggregate dynamics, eliminating the fixed-point problem that dominates standard heterogeneous-agent computation.

Definition 3 (Belief-invariant recursive competitive equilibrium (BRCE)).

A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, K, L, w, r)$ is a belief-invariant recursive competitive equilibrium if:

1. g_c, g_a, g_n, v solve the household's problem given arbitrary beliefs about aggregate Markov laws of motion for both endogenous and exogenous states.
2. K, L satisfy the production sector's optimality conditions.
3. Factor markets clear.
4. Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the policy function g_a and the exogenous idiosyncratic process to Φ .

Theorem 1 (Belief invariance).

In the BRCE, the law of motion Γ_Φ is unaffected by households' expectations about future aggregate states.

Proof. See Appendix A.6 of the Online Appendix. ■

The saving policy $a'(x; X) = \beta a((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)Z_L z \xi) + T(X)$ depends only on current prices and Z_L —not on forecasts. Under log utility, income and substitution effects of future return changes exactly offset, so all policies are belief-invariant and the aggregate law of motion inherits this property.

Proposition 4 (Unique existence of the BRCE).

The BRCE uniquely exists and the allocations are identical to the MRCE.

Proof. See Appendix A.7 of the Online Appendix. ■

Because the MRCE and BRCE coincide, the equilibrium is computed by forward simulation from any initial distribution Φ_0 and aggregate sequences $\{A_t, Z_{L,t}, \beta_t\}$. No fixed-point iteration is required, no assumption on expectations need be made, and no stationarity is imposed. Each period, the exact household policies determine individual decisions, aggregate moments are computed from the cross-section, and the distribution evolves forward. The same procedure applies whether aggregates follow stationary processes, deterministic trends, or both. This also enables the exact inversion: structural shocks can be recovered from data without taking a stand on expectation formation.

Remark 1 (Saving share under time-varying β).

Since β is an element of the aggregate state X , it may vary over time. With time-varying β , the value function takes the form $v(a, z, e; X) = \zeta \log a + B(z, e, X)$, where ζ satisfies $\zeta = (1 + \theta) + \beta \mathbb{E}[\zeta']$. Under belief invariance, households treat the current β as permanent, yielding $\zeta = (1 + \theta)/(1 - \beta)$ and saving share $s = \beta$. All propositions above then hold with the current β drawn from X : the saving policy is $a'(x; X) = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L \xi z) + T$, and the aggregate capital law of motion is $K' = \beta[(1 + \tilde{r} + \tilde{w}Z_L \xi)K + \tilde{w}Z_L \xi \text{cov}(z, a) + \tilde{r} \text{cov}(e, a)] + T$. No additional state variables or expectation formation are required. See Online Appendix A.8 for the ζ recursion derivation.

3.7 Macroeconomic implications

State-dependent aggregate dynamics The capital law of motion (17) decomposes into three channels: the representative-agent channel $(1 + \tilde{r} + \tilde{w}Z_L\xi)K$, the return-sorting channel $\tilde{r} \text{cov}(e, a)$, and the productivity-sorting channel $\tilde{w}Z_L\xi \text{cov}(z, a)$. These distributional terms make aggregate dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

Distributional channels The aggregate consumption function decomposes as:

$$C(\mathcal{M}, A, Z_L, \beta) = \underbrace{\frac{1-\beta}{1+\theta}(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \kappa\tilde{w}Z_L}_{\text{Representative-agent level}} + \underbrace{\frac{1-\beta}{1+\theta}(\tilde{r} \text{cov}(e, a) + \tilde{w}Z_L\xi \text{cov}(z, a))}_{\text{Distributional channel}} \quad (25)$$

The aggregate employment equation reveals a parallel decomposition:

$$L(\mathcal{M}, A, Z_L, \beta) = Z_L\kappa + \underbrace{\frac{Z_L\xi(1+\theta\beta)(K + \text{cov}(z, a))}{1+\theta}}_{\text{Leisure enhancement}} - \underbrace{\frac{\theta(1-\beta)[K(1+\tilde{r}) + \tilde{r} \overbrace{\text{cov}(e, a)}^{\text{Dist. channel}}]}{(1+\theta)\tilde{w}}}_{\text{Wealth effect}} \quad (26)$$

The wealth-effect term is amplified by $\text{cov}(e, a)$: wealthy households earning high returns consume more leisure, reducing labor supply. In periods of high return-wealth sorting, a positive productivity shock elicits a smaller employment response as the households with the largest wealth gains withdraw the most labor.

State-dependent inequality dynamics The K' equation (17) generates a “wealth spiral”: productivity shocks raise capital, which strengthens return-wealth sorting through $\text{cov}(e, a)$, which raises future capital. Discounting ($\beta < 1$) and the transfer T counterbalance. The persistence of the sorting mechanism—governed by the near-unit-root dynamics of the wealth distribution—controls the speed of concentration (Appendix B.4).

3.8 Asset pricing implications

Individual return heterogeneity, traditionally viewed as diversifiable, affects aggregate prices once it is sorted with wealth (Constantinides and Duffie, 1996; Storesletten et al., 2004).

Households with persistently high returns accumulate disproportionate wealth, shifting labor supply and equilibrium returns:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \text{cov}(e, a)} < 0 \quad (27)$$

Higher return-wealth sorting reduces the equilibrium interest rate through two channels: a static one (the wealth effect lowers labor supply, raising the capital-labor ratio) and a dynamic one (persistent sorting accelerates capital accumulation, depressing future returns). The full decomposition and quantitative implications appear in Appendix C.

4 Quantitative Analysis

The quantitative exercise uses the analytical structure for two purposes. First, it asks whether the inversion recovers economically interpretable shocks rather than simply relabeling three observables. Second, it measures the distributional terms that the theory identifies as missing from representative-agent aggregation. It does not use exact fit as evidence: with three observables and three recovered aggregate shocks, output, capital, and hours are matched mechanically. The tests are instead non-targeted: simulated Krusell–Smith economies, external labor and saving indicators, forecast performance, and distributional implications.

4.1 Inversion and Calibration

The application uses annual U.S. data from 1950 to 2019 on real per-capita GDP, the per-capita capital stock, and aggregate hours. The aggregate equations are the production

function, labor supply, and capital accumulation:

$$Y_t = A_t K_{eff,t}^\alpha L_t^{1-\alpha}, \quad (28)$$

$$L_t = Z_{L,t} \kappa + \frac{Z_{L,t} \tilde{\xi}_t (1 + \theta \beta_t)}{1 + \theta} (K_t + \text{cov}_t(z, a)) - \frac{\theta(1 - \beta_t)}{(1 + \theta) \tilde{w}_t} (K_t(1 + \tilde{r}_t) + \tilde{r}_t \text{cov}_t(e, a)), \quad (29)$$

$$K_{t+1} = \beta_t \left[(1 + \tilde{r}_t + \tilde{w}_t Z_{L,t} \tilde{\xi}_t) K_t + \tilde{r}_t \text{cov}_t(e, a) + \tilde{w}_t Z_{L,t} \tilde{\xi}_t \text{cov}_t(z, a) \right] + T_t, \quad (30)$$

where $\tilde{\xi}_t \equiv \xi / A_t^{1/(1-\alpha)}$ is the growth-adjusted complementarity coefficient (Online Appendix B). Note that equation (29) is implicit in L because prices depend on L through the marginal products (Online Appendix A proves unique existence). The distributional covariances entering $K_{eff,t}$ are obtained from a preliminary two-series inversion that provides the moment-recursion covariance path (Online Appendix E). Given the distributional state, these equations recover $(A_t, \beta_t, Z_{L,t})$ period by period. TFP is identified from production, the saving-margin wedge β_t from capital accumulation, and the labor-efficiency wedge $Z_{L,t}$ from labor supply. The distribution is then simulated forward using the exact household policy.

The recovered shocks have distinct structural interpretations. A_t captures neutral total factor productivity. β_t captures time-varying patience—or equivalently, demand-side forces operating through the saving margin. $Z_{L,t}$ is a model primitive (Section 2), not a reduced-form residual. Three features distinguish it from the labor wedge of [Chari et al. \(2007\)](#). First, it has a direct structural interpretation: time variation maps to shifts in labor force composition, matching quality, or utilization intensity. Second, $Z_{L,t}$ scales household decisions but not the production function, making it separately identified from TFP. Third, it has distributional consequences: $Z_{L,t}$ enters the saving policy and equilibrium prices, so shifts in labor productivity alter the dynamics of wealth inequality.

The calibration is deliberately conventional. The capital share α_t and depreciation rate δ_t are time-varying, drawn from Penn World Tables and BEA fixed-assets data respectively. The analytical expressions carry over with α replaced by α_t period by period, since each period's equilibrium is contemporaneous. Preference parameters are standard: $\beta = 0.96$ is

the reference annual discount factor, and the leisure weight is $\theta = 4$ (Online Appendix E.12 reports robustness to alternative values). Idiosyncratic earnings risk follows [Storesletten et al. \(2004\)](#), and idiosyncratic return risk follows [Fagereng et al. \(2020\)](#). The transfer share is calibrated to place steady-state wealth concentration near the U.S. range.

Table 1: Calibrated parameters

Parameters	Description	Value	Source
Data-anchored (time-varying)			
α_t	Capital share	0.351–0.414	Penn World Tables
δ_t	Depreciation rate	0.037–0.058	BEA Fixed Assets
A_t	Aggregate TFP	Inverted	GDP, capital, hours
β_t	Discount factor	Inverted	Capital accumulation
$Z_{L,t}$	Labor productivity	Inverted	Labor supply
Calibrated (fixed)			
β	Reference discount factor	0.9600	Annual standard
θ	Labor disutility	4.0000	KS98 validation
κ	Common time endowment	0.2700	$\kappa < \min(L_{\text{data}})$
ξ	Wealth–leisure complementarity	0.0213	Saving/labor income ≈ 0.33
τ	Proportional income tax rate	0.3000	Effective tax rate
η	Transfer share of tax revenue	0.0133	Wealth Gini ≈ 0.80
Idiosyncratic risk			
ρ_z	Prod. persistence	0.9500	Storesletten et al. (2004)
σ_z	Prod. volatility	0.1200	Storesletten et al. (2004)
ρ_e	Return persistence	0.9990	Fagereng et al. (2020)
σ_e	Return volatility	0.0100	Fagereng et al. (2020)

Notes: Calibration is annual. The table reports the baseline used in the U.S. inversion and in the distributional simulation.

4.2 Validation and Recovered Shocks

Krusell–Smith validation The first check is whether the inversion spuriously creates additional shocks when the data are generated by a model with only aggregate TFP risk. We simulate three Krusell–Smith variants and apply the same inversion to simulated output, capital, and hours. Figure 1 and Table 2 show that the recovered TFP matches the true process, while β_t and $Z_{L,t}$ remain essentially constant. Two extensions, one with endogenous

labor supply and one with Frisch labor supply and convex adjustment costs, deliver the same result. The inversion therefore does not mechanically manufacture preference or labor shocks when the true data-generating process has only aggregate TFP.

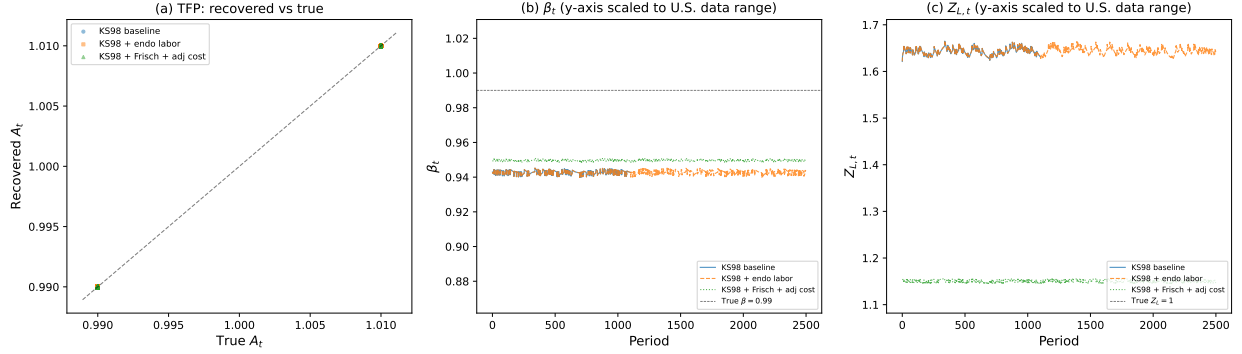


Figure 1: Krusell–Smith validation: recovered shocks from simulated data

Notes: Sequential inversion applied to simulated (Y_t, K_t, L_t) from three Krusell–Smith variants. The recovered TFP aligns with the true aggregate TFP process, while the recovered β_t and $Z_{L,t}$ series are nearly flat on the U.S. data scale.

Applying the same inversion to U.S. data reveals economically significant departures from the Krusell–Smith benchmark. The discount factor β_t fluctuates more than in the simulated economies, but the labor-efficiency wedge is the dominant departure. U.S. capital accumulation is well approximated by a single TFP shock with near-constant preferences, while labor-market fluctuations require an additional structural margin that a one-shock Krusell–Smith economy cannot generate.

Table 2: Krusell–Smith validation: inversion accuracy

	$\text{mean} A^{\text{inv}} - A^{\text{true}} /A$	$\text{CV}(\beta)$	$\sigma(\Delta \log Z_L)$	$\text{mean} Y^{\text{inv}} - Y^{\text{true}} /Y$
KS98 baseline	7.5×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + endo labor	7.3×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + Frisch + adj cost	2.5×10^{-5}	0.05%	0.18%	$< 10^{-15}$
U.S. data (1950–2019)	—	1.78%	3.42%	exact

Notes: In the simulated economies, β and Z_L are constant by construction and only aggregate TFP varies. The final row applies the same inversion to U.S. data.

Recovered series Figure 2 reports the recovered structural series. TFP grows from 0.807 in 1950 to 1.79 in 2019, tracing the postwar expansion, the 1970s slowdown, the IT revolution, and post-2008 stagnation. The saving-margin wedge β_t is persistent and countercyclical, with a secular decline from 1950 to 2019. The labor-efficiency wedge $Z_{L,t}$ fluctuates at business-cycle frequencies and captures movements in effective labor input beyond measured hours.

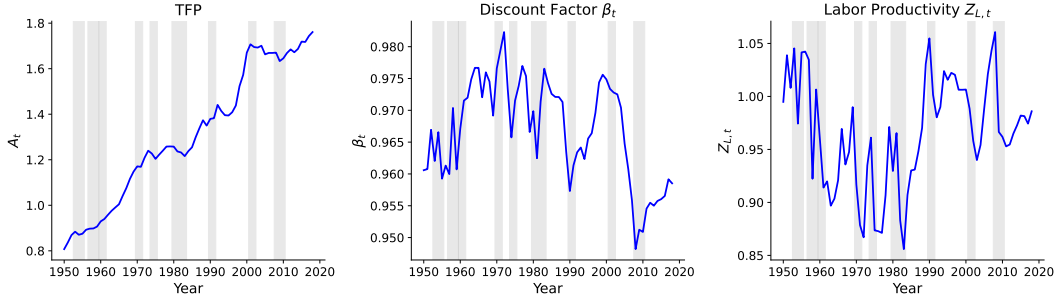


Figure 2: Recovered structural series from the data-anchored inversion

Notes: Top: aggregate TFP. Middle: saving-margin wedge β_t . Bottom: labor-efficiency wedge $Z_{L,t}$. The inversion uses observed output, capital, and hours. Shaded areas indicate NBER recession dates.

The MRCE-implied TFP tracks the representative-agent analogue closely at business cycle frequencies and correlates strongly with the Fernald utilization-adjusted series (Online Appendix E.1).

External labor discipline The second check asks whether the recovered labor-efficiency wedge is related to external labor-market margins not used in the inversion. At business-cycle frequencies, $Z_{L,t}$ is strongly correlated with the employment-population ratio and labor-force participation, the extensive margins it is designed to capture. The BLS labor composition index, by contrast, has the opposite cyclical sign: in recessions, low-productivity workers are disproportionately laid off, raising measured average quality, while $Z_{L,t}$ falls because utilization and matching deteriorate. The sign reversal supports the interpretation of Z_L as a utilization and matching margin rather than a demographic composition index.

Saving-margin discipline The declining trend in β_t is not used as a target, so it needs external discipline. Figure 3 compares the recovered saving-margin wedge with three outside series: the personal saving rate, labor-force participation, and revolving consumer credit relative to GDP. The correlations are not a structural proof, but they help interpret β_t

Table 3: External validation: recovered $Z_{L,t}$ and labor-market indicators

External series	Level corr	HP-cycle corr
Employment-population ratio (EMRATIO)	0.48	0.81
Labor force participation rate (CIVPART)	0.28	0.54
Avg. weekly hours, manufacturing (AWHMAN)	0.42	0.41

Notes: The external series are not used in the inversion. HP-cycle correlations use annual data with $\lambda = 6.25$.

as a compact saving-side wedge that absorbs forces operating through the saving margin, including demographic shifts, financial liberalization, and precautionary motives.

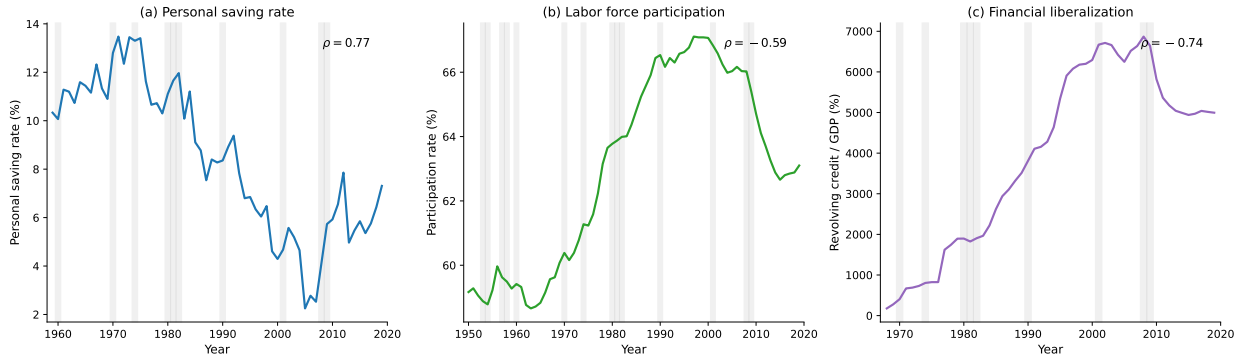


Figure 3: External discipline for the saving-margin wedge

Notes: External indicators from FRED, not used in the inversion. Panels show the U.S. personal saving rate, civilian labor-force participation, and revolving consumer credit as a share of GDP against the recovered β_t . Shaded areas indicate NBER recession dates.

A two-dimensional business cycle Figure 4 displays pairwise scatter plots of the recovered shocks. The most striking finding is the near-orthogonality of TFP with both $Z_{L,t}$ and β_t : technology evolves independently of the labor-saving block. The key cyclical correlation is between labor efficiency and patience, $\rho(\Delta \log Z_{L,t}, \Delta \beta_t) = -0.858$. Labor-market deterioration is systematically accompanied by higher saving, consistent with a precautionary channel. Together with the Krusell–Smith validation, the scatter plot shows that the inversion is not merely relabeling one residual: trend productivity and the labor-saving block occupy different dimensions of the data.

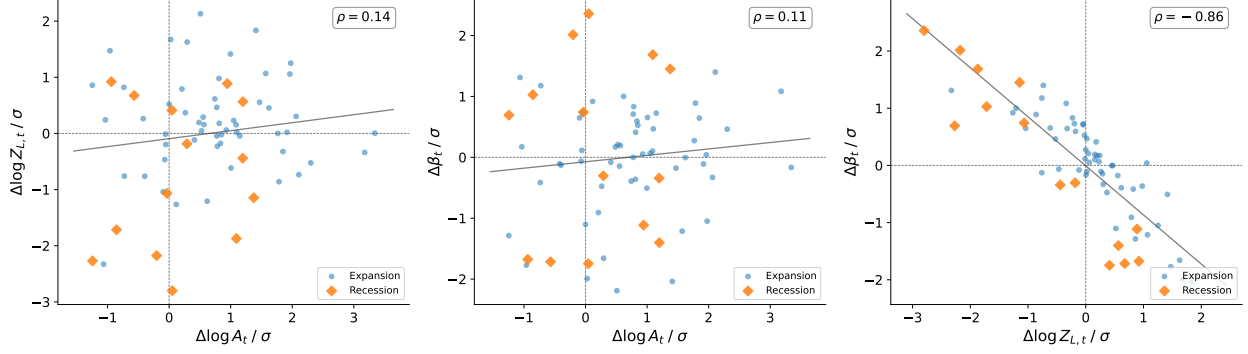


Figure 4: Pairwise scatter of recovered shock changes: the two-dimensional business cycle

Notes: Each point is one year. Blue circles are expansion years; orange diamonds are recession years. Changes are normalized by their unconditional standard deviations. The right panel shows the labor-saving tandem; the left and center panels show that TFP innovations are nearly orthogonal to the other two shocks.

4.3 Conditional Forecasting

A distinctive advantage of the MRCE is that conditional forecasting is computationally direct. In conventional heterogeneous-agent models, forecasting requires re-solving the household dynamic program or estimating a perceived aggregate law of motion, because policy functions depend on beliefs about future aggregate states. In the MRCE, household policies are backward-looking and linear in wealth. Given any future sequence for $(A_t, \beta_t, Z_{L,t})$, the model produces output, capital, labor, prices, consumption, investment, and the distributional state by forward simulation alone.

Rolling forecasts We exploit this property with a rolling forecast exercise. For each origin year $t \in \{1965, \dots, 2009\}$, the inversion recovers structural shocks from data available up to t , and the model is simulated forward holding shocks at their last recovered values. The resulting forecast jointly produces output, capital, labor, prices, consumption, investment, and the distributional state, all satisfying the production function, capital accumulation, and market clearing simultaneously.

Figure 5 compares the MRCE forecast to a recursive AR benchmark that forecasts the structural inputs using the same information set and aggregates them through the same production function. At both one- and three-year horizons, the MRCE has lower RMSE for output (3.1% vs. 7.0% at $h = 1$; 5.1% vs. 9.4% at $h = 3$) and labor (2.6% vs. 4.2%; 5.4% vs. 6.1%). For capital, the AR benchmark exploits the high persistence of the raw capital stock

($\rho = 0.99$) and is difficult to beat at short horizons. The MRCE’s advantage is structural: it produces a jointly consistent forecast of all prices and quantities, while the AR inputs evolve independently and are not required to satisfy any equilibrium condition.

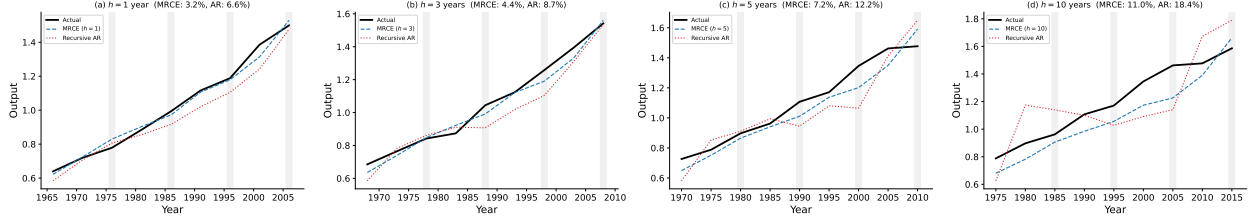


Figure 5: Rolling forecast errors by horizon

Notes: For each origin year, the model is simulated forward holding recovered shocks at their last values. The recursive AR benchmark forecasts structural inputs using the same information set and aggregates through the same production function. Panel titles report RMSEs. Shaded areas indicate NBER recession dates.

4.4 Aggregate Dynamics

The analytical aggregate equations imply three quantitative lessons. First, representative-agent aggregation is an excellent approximation in the baseline calibration. Investment decomposes into a representative-agent channel, a return-sorting channel, and a productivity-sorting channel; the representative-agent channel accounts for 93.2 percent of mean investment. This explains why the Krusell–Smith approximation works so well even though the exact state contains additional covariances.

Second, the omitted covariances still create a state-dependent amplification channel. We compute generalized impulse responses to a 2 percent negative TFP shock from the 1952 distribution, when wealth inequality is low, and from the 2019 distribution, when wealth inequality is high. The impact effect is 3 percent larger from the 2019 state (-2.29% versus -2.22%), and the cumulative 10-year output loss is modestly larger (-20.4% versus -20.2%). The magnitude is small at the conservative baseline calibration, but the sufficient statistic is transparent: amplification rises with return-wealth sorting.

Third, the direct asset-pricing effect of return-wealth sorting is negligible at $\sigma_e = 0.01$. The interest-rate effect is approximately -0.001 basis points. This is a useful negative result. In the baseline calibration, return heterogeneity matters more as an amplification state variable than as a quantitatively important risk-premium channel. An Epstein–Zin

extension, reported in the Online Appendix, is a more promising route for asset-pricing applications.

4.5 Distributional Dynamics and the Sorting Accelerator

TFP growth and inequality The central distributional mechanism is a sorting accelerator. TFP growth raises returns, persistent return heterogeneity causes high-return households to become increasingly wealthy, and wealth concentration then raises the importance of return-wealth sorting in future dynamics. The relevant covariance, $\text{cov}(e, a)$, is therefore highly persistent: the dominant eigenvalue governing distributional dynamics exceeds one in 100 percent of sample years, with mean 1.02.

Four distributional counterfactuals isolate the mechanism. The baseline simulation uses the historical paths of all recovered shocks. The TFP-only counterfactual keeps only the TFP path. The fixed- A counterfactual removes TFP growth. The all-fixed counterfactual holds all aggregate shocks fixed. Figure 6 reports the results. Both sorting channels grow substantially along the baseline path, but for different reasons. Return-wealth sorting grows through near-unit-root persistence, while productivity-wealth sorting grows through recession-driven bursts in idiosyncratic dispersion. Without TFP growth, return-wealth sorting stagnates; without countercyclical dispersion, productivity sorting loses its recession spikes.

Analytical decomposition The MRCE makes the sorting accelerator precise. In the return-sorting channel, persistence is governed by $\phi_e \rho_e (1 + \tilde{r})$, which is close to $1/\beta$ because return shocks are highly persistent. In the productivity-sorting channel, persistence is lower, but countercyclical dispersion raises the innovation rate during downturns. Figure 7 decomposes these forces. The secular component comes from persistent return heterogeneity; the cyclical component comes from recession-driven increases in idiosyncratic dispersion.

Who gains from TFP growth? Table 4 reports the impact of 1 percent TFP growth by wealth percentile. The top decile captures 81.6 percent of the aggregate wealth gain, exceeding its 77.2 percent wealth share, because the return-sorting and productivity-sorting channels compound. The top 1 percent alone captures 68.1 percent. The table is not a

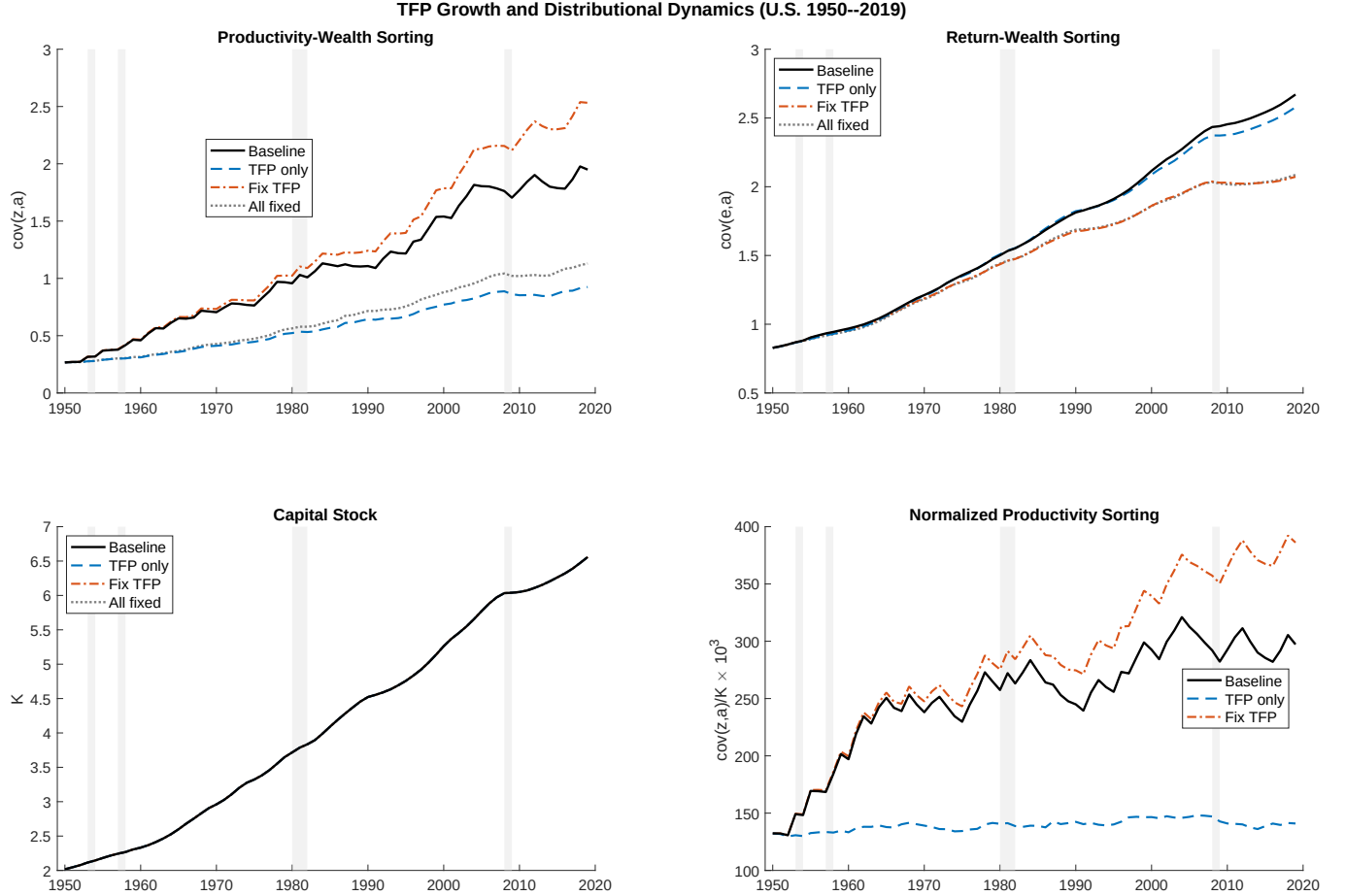


Figure 6: TFP growth and distributional dynamics

Notes: Top left: productivity-wealth covariance $\text{cov}(z, a)$ under four counterfactual scenarios. Top right: return-wealth covariance $\text{cov}(e, a)$. Bottom left: aggregate capital. Bottom right: normalized sorting intensity $\text{cov}(z, a)/K$. Shaded areas indicate NBER recession dates.

tax counterfactual; it is a sufficient-statistic calculation showing how the analytical channels allocate a common productivity improvement across the wealth distribution.

Historical gain shares The steady-state calculation is conservative because it holds the distribution fixed. Along the historical path, the distribution itself changes. Table 5 reports decade averages of TFP-driven wealth gains by wealth decile, and Figure 8 visualizes the evolving distribution. The top decile captures approximately 67.3 percent of TFP-driven wealth gains in the 1950s, rising to 69.4 percent by the 2010s. This increase reflects the endogenous accumulation of return-wealth sorting. As the wealth distribution concentrates through the Kesten mechanism, the return-sorting and productivity-sorting channels com-

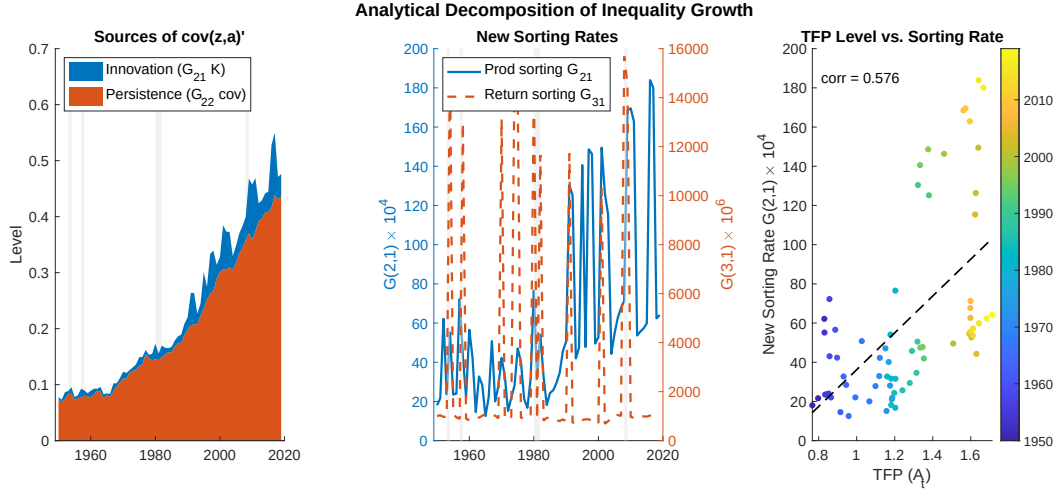


Figure 7: Analytical decomposition of distributional sorting

Notes: Left: innovation and persistence contributions to $\text{cov}(z, a)'$. Center: sorting-rate time paths for productivity and return shocks. Right: TFP level against productivity sorting. Shaded areas indicate NBER recession dates.

Table 4: Distributional impact of 1% TFP growth at steady state

Group	Wealth	Gain	Mean Gain	Channel (%)		
	Share (%)	Share (%)	(%)	Capital	Labor	Transfer
Bottom 10%	1.2	0.8	0.0806	65.1	18.9	16.0
10–20%	1.4	1.0	0.0850	67.9	19.8	12.3
20–30%	1.7	1.2	0.0873	69.4	20.2	10.4
30–40%	1.9	1.4	0.0893	70.6	20.5	8.9
40–50%	2.2	1.7	0.0913	71.7	20.6	7.6
50–60%	2.5	2.0	0.0932	72.8	20.7	6.5
60–70%	3.0	2.4	0.0954	74.0	20.7	5.3
70–80%	3.7	3.1	0.0982	75.4	20.5	4.1
80–90%	5.3	4.6	0.1019	77.2	20.1	2.8
Top 10%	77.2	81.6	0.1102	84.0	15.9	0.2
Top 5%	72.8	77.6	0.1138	84.2	15.7	0.1
Top 1%	63.1	68.1	0.1195	84.6	15.4	0.0

Notes: Gains computed from 1% TFP growth at the stationary distribution of simulated households. “Capital” is the gain from higher after-tax capital income; “Labor” is the gain from higher labor income; “Transfer” is the common transfer component.

pound through the exact saving rule $a' = \beta R_i a + T$, and the top decile’s share of any marginal productivity improvement grows accordingly.

The cumulative effect of the sorting accelerator is most visible in counterfactuals involving persistent shocks. The next subsection uses two historical capital-tax reforms to show

Table 5: Share of TFP-driven wealth gains by wealth decile (%), decade averages

Decade	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Top 5%	Top 1%
1950s	1.3	1.7	2.1	2.4	2.9	3.5	4.4	5.7	8.8	67.3	59.7	44.4
1960s	1.3	1.7	2.1	2.5	2.9	3.5	4.4	5.8	8.8	67.1	59.4	44.1
1970s	1.2	1.7	2.0	2.4	2.9	3.5	4.4	5.8	8.9	67.1	59.4	44.0
1980s	1.2	1.6	2.0	2.4	2.9	3.5	4.4	5.8	8.9	67.3	59.5	43.8
1990s	1.1	1.6	1.9	2.3	2.8	3.4	4.3	5.8	8.9	67.8	59.9	44.0
2000s	1.0	1.5	1.8	2.2	2.7	3.3	4.3	5.8	9.0	68.4	60.3	43.8
2010s	0.9	1.3	1.7	2.1	2.6	3.2	4.1	5.7	9.0	69.4	61.2	44.3
Full sample	1.1	1.6	1.9	2.3	2.8	3.4	4.3	5.8	8.9	67.8	59.9	44.0

Notes: Each entry reports the average share of aggregate TFP-driven wealth gains captured by the indicated wealth decile. Gains are computed period by period along the historical simulation.

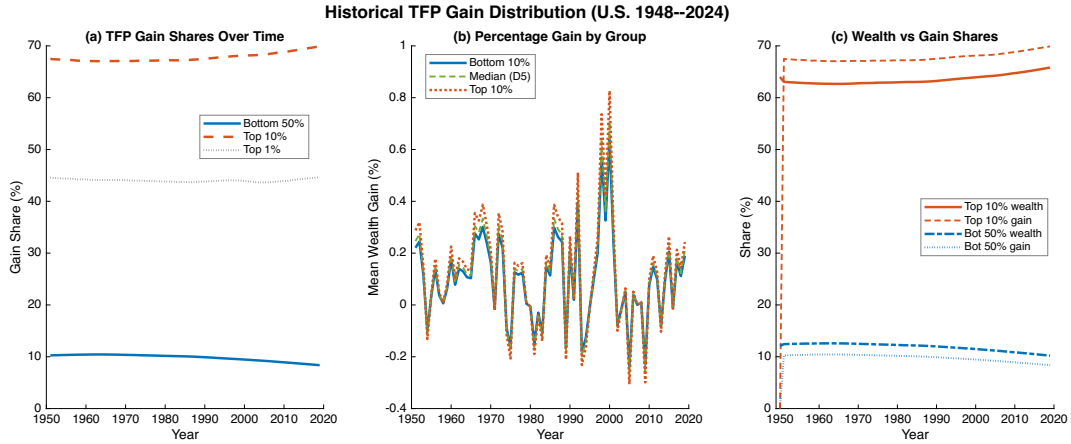


Figure 8: Historical distribution of TFP-driven gains

Notes: Period-by-period decomposition of the share of TFP-driven wealth gains by wealth percentile, U.S. 1950–2019.

that the near-unit-root sorting dynamics prevent mean reversion after a policy change, so distributional gaps opened by even transient reforms compound for decades.

4.6 Distributional impact of tax reforms

The near-unit-root persistence of wealth sorting implies that even temporary changes in capital taxation can produce almost permanent shifts in the wealth distribution. We test this prediction using four historically abrupt capital tax changes from both directions of the political spectrum: two cuts (Reagan ERTA 1981, τ_K from 0.380 to 0.375; Bush EGTRRA 2001, τ_K from 0.395 to 0.355) and two increases (OBRA 1993 under Clinton, τ_K from 0.340

to 0.360; ATRA 2013 under Obama, τ_K from 0.278 to 0.310).²

For each episode, two full agent-based simulations (500,000 agents) use identical aggregate shocks and idiosyncratic histories. The counterfactual is *reform-only*: $\tau_{K,t}^{\text{cf}}$ equals the factual path minus the named reform’s reform-year shock from the reform date onward, leaving all other reforms intact. For example, the OBRA-only counterfactual subtracts +0.020 from $\tau_{K,t}$ for $t \geq 1993$ but keeps the Bush 2001 cut, ATRA 2013 increase, and TCJA 2017 cut in both factual and counterfactual paths. This isolates the marginal effect of each named reform rather than the cumulative effect of “no policy change after the reform date.” Figures 9 and 10 report the cuts and increases respectively, in the same panel layout.

Several features of Figure 9 merit attention. First, the Gini gap opens immediately at each reform date and continues to widen for decades, confirming the near-unit-root prediction: once the wealth distribution is pushed toward greater concentration, persistence compounds the initial shift rather than allowing mean reversion. Second, the wage panel (row 4) reveals the growth side of the tradeoff: lower capital taxes stimulate capital accumulation, raising the marginal product of labor and hence equilibrium wages—all households benefit from higher wages even as inequality rises. Third, the return-wealth covariance (row 5) diverges at each reform date, tracing the sorting mechanism directly: lower τ_K raises after-tax capital returns, which disproportionately benefit high-wealth households, accelerating $\text{cov}(e, a)$ growth.

Figure 10 confirms that the same mechanism runs in reverse for tax increases. After OBRA 1993 raises τ_K by +0.020 (from 0.340 to 0.360), the factual wealth Gini falls relative to the reform-only counterfactual, the return-wealth covariance compresses, and output and wages contract. The Bush (−0.040) and ATRA (+0.032) reform-year shocks dominate the Reagan (−0.005) and OBRA (+0.020) shocks in magnitude, but cumulative distributional effects are governed jointly by shock size and post-reform time horizon: OBRA 1993 has 26 post-reform years in the sample, ATRA 2013 only 6. The reform-only design isolates each reform from all subsequent policy: in particular, the OBRA panel is no longer contaminated

²Annual effective tax rates ($\tau_{L,t}, \tau_{K,t}$) used in the inversion and counterfactuals are not directly observed at annual frequency. We construct them as a year-by-year stylized series consistent with the decade averages of [McDaniel \(2007\)](#) and the timing of major federal legislation (Korean War surtax, Kennedy 1964, ERTA 1981, TRA 1986, OBRA 1993, EGTRRA 2001, JGTRRA 2003, ARRA 2009, ATRA 2013, TCJA 2017). The full series is in `exact_ge_mapping/data/effective_tax_rates.csv`; see `build_tax_data.py` for the construction. Counterfactual conclusions about *differences* between factual and reform-only paths are robust to reasonable variation in the tax-rate series because both scenarios are subject to the same construction.

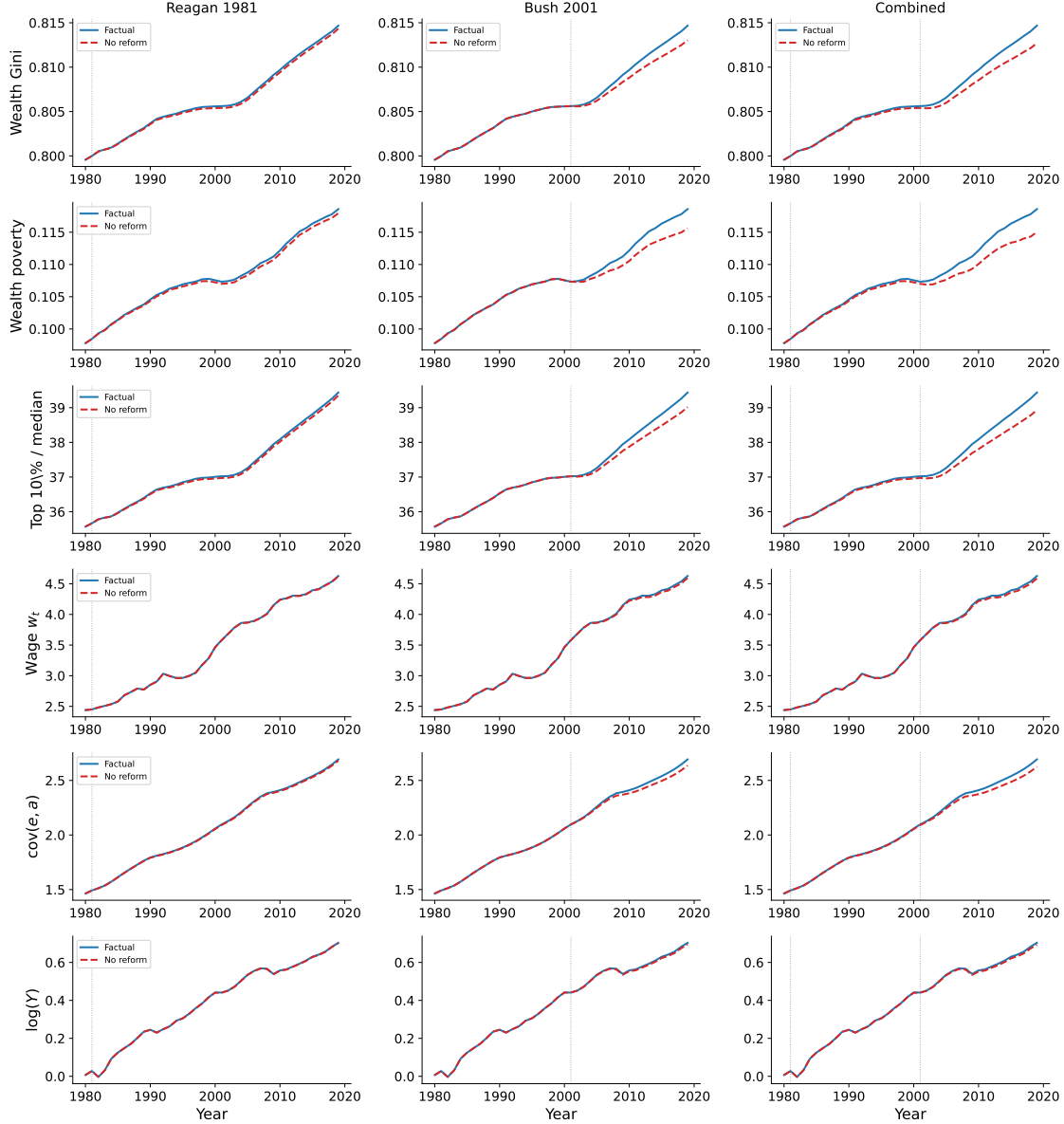


Figure 9: Macroeconomic impact of capital tax cuts (Reagan 1981, Bush 2001)

Notes: 500,000-agent simulation, 1980–2019. Columns: Reagan 1981, Bush 2001, and combined cuts. Rows show wealth Gini, wealth poverty (share of households with wealth below half the median), top 10%/median ratio, wage, return-wealth covariance, and output. Solid blue: factual; dashed red: counterfactual (no reform). Capital evolves endogenously; all GE prices re-solved each period.

by the post-1993 Bush 2001 and TCJA 2017 cuts.

Panel A of Table 6 quantifies the growth-versus-inequality tradeoff for the cuts under the reform-only counterfactual. Stacking both reforms, the combined isolated effect by 2019 is a 1.01% output gain and 0.87% wage increase, offset by a 0.20 percentage point wealth Gini increase, a 1.30% rise in the top 10%/median ratio, and a 3.11% increase in the wealth-

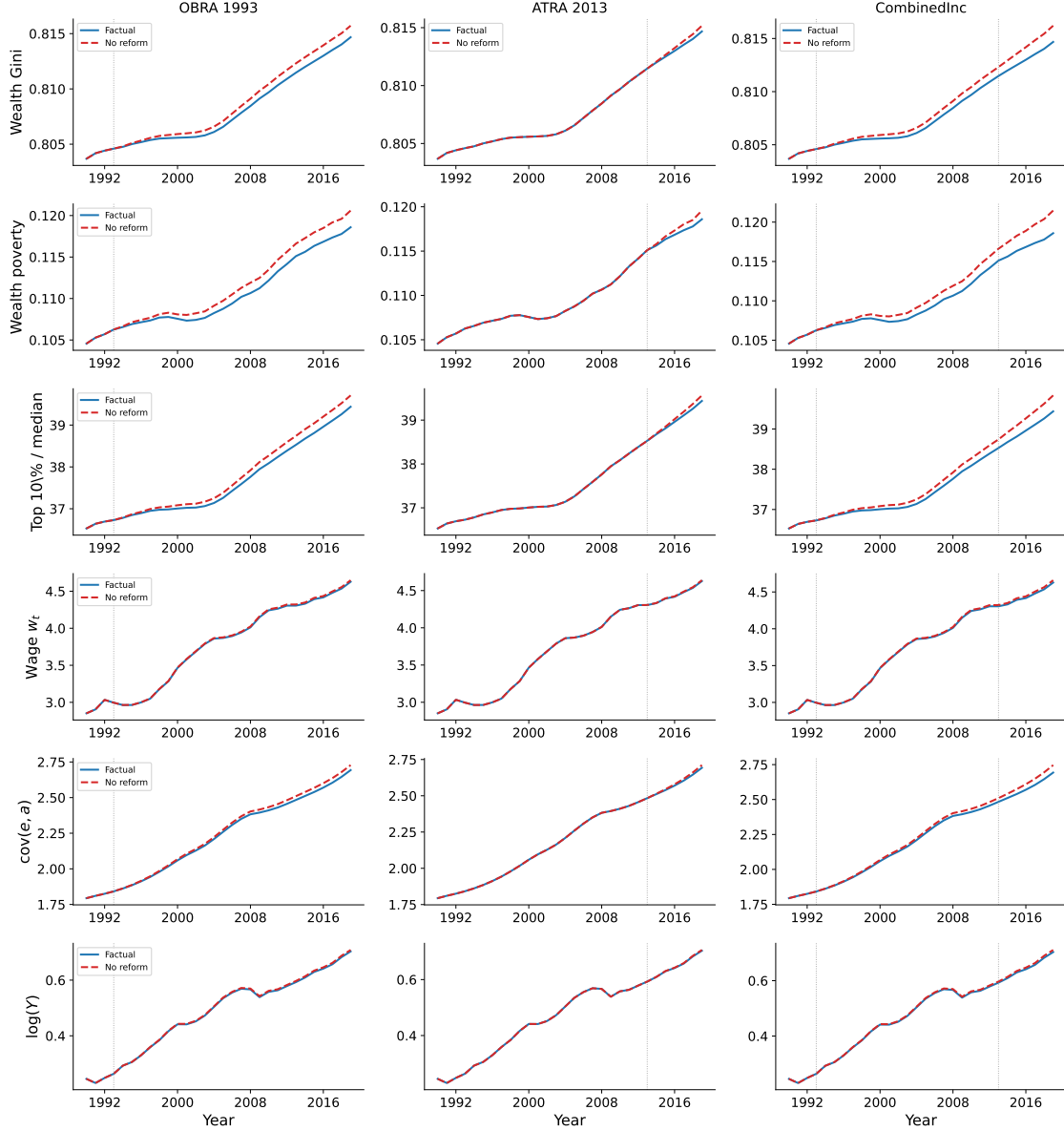


Figure 10: Macroeconomic impact of capital tax increases (OBRA 1993, ATRA 2013)

Notes: Same simulation and panel layout as Figure 9. Columns: OBRA 1993, ATRA 2013, and combined increases. Solid blue: factual; dashed red: reform-only counterfactual (subtract only the named reform's reform-year τ_K shock from the factual path going forward, keeping all subsequent reforms intact). The horizontal axis starts at 1990 to focus on the post-reform window.

poverty rate. The income share of the top 1% rises by 0.40 percentage points while the bottom 50% loses 0.18 points. The magnitudes are smaller than under the freeze-pre-reform design (which conflates each reform with all subsequent policy), but the signs and rankings are the same: Bush EGTRRA delivers most of the cumulative effect because its reform-year cut (-0.040) is much larger than Reagan ERTA's (-0.005). Even under the cleaner

interpretation, the bottom decile gains in absolute income levels but loses share, as the sorting mechanism channels the growth dividend toward the top.

Panel B of Table 6 reports the symmetric numbers for the two tax-increase episodes under the same reform-only design. Because OBRA 1993 lies 17 years before 2010 and ATRA 2013 lies 3 years *after* 2010, we use 2002 (9 years post-OBRA, before ATRA) and 2019 (terminal) as the horizons. Signs are uniformly negative for output, wage, and inequality measures, mirroring the cuts. By 2019 the combined isolated effect of both increases is -0.77% output, -0.67% wage, -0.15 pp wealth Gini, -1.00% top 10%/median, and -2.37% wealth poverty. The income share of the top 1% falls by 0.48 percentage points and the bottom 50% gains 0.21 points: the same sorting mechanism runs in reverse, redistributing toward the bottom when capital is taxed more heavily. OBRA 1993 carries most of the cumulative effect because it has 26 post-reform years in the sample versus only 6 for ATRA 2013.

The exercise establishes that the macro side’s tax counterfactuals are not skewed toward one political direction: the same near-unit-root sorting mechanism produces decades-long distributional drift after both major cuts and major increases, with appropriately signed effects. Read together, the 1981, 1993, 2001, and 2013 episodes span the full postwar range of capital tax policy, and each leaves a structural footprint that the analytical framework traces directly.

4.7 Cross-Sectional Dynamics

The same historical simulation generates cross-sectional moments. Table 7 compares the model’s wealth Gini with SCF-based estimates at survey years. The model captures the broad level of U.S. wealth inequality through the Kesten-process rule $a' = \beta R_i a + T$, with persistent return shocks and positive transfers. It rises more smoothly than the data. This is a useful miss: time-varying return dispersion, top income growth, and housing-market divergence are natural candidates for the remaining trend.

Figure 11 summarizes the time-series path of wealth, consumption, and income inequality. Three patterns stand out. First, wealth concentration rises monotonically from 0.827 to 0.876. The driver is the near-unit-root persistence of return-wealth sorting—the same Kesten mechanism that generates the sorting accelerator. Second, consumption inequality also rises

Table 6: Growth vs. inequality: capital tax reforms (reform-only counterfactuals)

<i>Panel A: Tax cuts</i>						
	Reagan 1981		Bush 2001		Combined	
Δ	2010	2019	2010	2019	2010	2019
<i>Growth (%)</i>						
Output	+0.13	+0.16	+0.44	+0.84	+0.58	+1.01
Wage	+0.12	+0.14	+0.42	+0.73	+0.54	+0.87
<i>Inequality measures</i>						
Wealth Gini (pp)	+0.03	+0.03	+0.09	+0.16	+0.12	+0.20
Top 10%/median (%)	+0.18	+0.22	+0.56	+1.07	+0.74	+1.30
Wealth poverty (%)	+0.46	+0.55	+1.47	+2.59	+1.90	+3.11
<i>Income share shift (pp)</i>						
Top 1%	+0.04	+0.04	+0.41	+0.36	+0.45	+0.40
Top 10%	+0.04	+0.04	+0.38	+0.35	+0.42	+0.39
Bottom 50%	−0.02	−0.02	−0.17	−0.16	−0.18	−0.18
Bottom 10%	−0.00	−0.00	−0.02	−0.02	−0.02	−0.02
<i>Panel B: Tax increases</i>						
	OBRA 1993		ATRA 2013		Combined	
Δ	2002	2019	2002	2019	2002	2019
<i>Growth (%)</i>						
Output	−0.19	−0.51	0.00	−0.26	−0.19	−0.77
Wage	−0.15	−0.44	0.00	−0.24	−0.15	−0.67
<i>Inequality measures</i>						
Wealth Gini (pp)	−0.04	−0.10	0.00	−0.05	−0.04	−0.15
Top 10%/median (%)	−0.25	−0.68	0.00	−0.32	−0.25	−1.00
Wealth poverty (%)	−0.74	−1.67	0.00	−0.82	−0.74	−2.37
<i>Income share shift (pp)</i>						
Top 1%	−0.14	−0.16	0.00	−0.33	−0.14	−0.48
Top 10%	−0.14	−0.16	0.00	−0.31	−0.14	−0.47
Bottom 50%	+0.06	+0.07	0.00	+0.14	+0.06	+0.21
Bottom 10%	+0.01	+0.01	0.00	+0.02	+0.01	+0.03

Notes: Reform-only counterfactuals: each CF subtracts only the named reform’s reform-year τ_K shock from the factual path going forward, leaving all other reforms intact. Reform-year shocks: Reagan -0.005 , Bush -0.040 , OBRA $+0.020$, ATRA $+0.032$. Growth and “Top 10%/median” / “Wealth poverty” rows: (factual $-$ counterfactual) / counterfactual $\times 100$. “Wealth Gini” row: percentage-point shift in Gini coefficient $\times 100$. “Wealth poverty” is the fraction of households with wealth below half the median. Income share shifts in percentage points. “Combined” stacks the panel’s two reform shocks additively. ATRA 2013 entries at 2002 are zero because the reform has not yet occurred.

(from 0.281 to 0.356). The consumption rule $c_i \propto (1 - \beta)R_i a_i$ loads directly on wealth, so cross-sectional dispersion is amplified as the wealth distribution concentrates. Third, income

inequality is flatter (mean Gini 0.404). Two offsetting forces are at work: rising wealth inequality pushes capital income apart, but the labor component—which scales with $Z_{L,t}z_i$ rather than wealth—compresses during recessions when low-productivity workers exit. The divergence between consumption and income Gini is a testable implication of the MRCE’s exact policy structure.

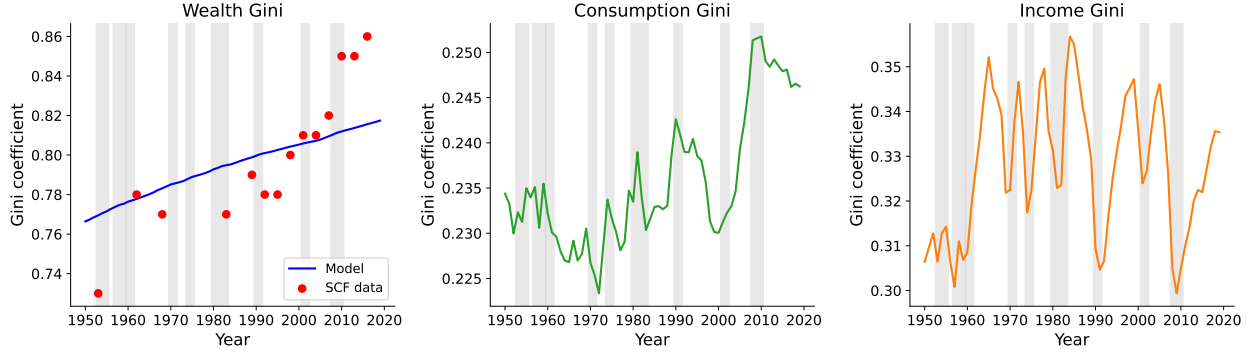


Figure 11: Historical cross-sectional distributional dynamics

Notes: Historical simulation using the recovered aggregate shocks and exact household policies. The wealth panel includes SCF-based estimates as external benchmarks. Shaded areas indicate NBER recession dates.

Table 7: Wealth Gini: model vs. data

Year	Data	Model	Year	Data	Model
1953	0.73	0.830	1998	0.80	0.864
1962	0.78	0.838	2001	0.81	0.865
1968	0.77	0.843	2004	0.81	0.867
1983	0.77	0.855	2007	0.82	0.869
1989	0.79	0.858	2010	0.85	0.871
1992	0.78	0.860	2013	0.85	0.873
1995	0.78	0.862	2016	0.86	0.875

Notes: Data are SCF-based estimates reported by [Kuhn and Ríos-Rull \(2025\)](#). Model moments are from the historical simulation using exact household policies.

The time-series Gini compresses the distribution into one number. Figure 12 overlays the full cross-sectional wealth distribution (normalized by mean wealth) in 1980 and 2015. The 2015 distribution exhibits a fatter right tail and a thinner middle: the mass between 0.5 and 1.5 times mean wealth declines while the upper tail extends. This is the distributional signature of the sorting accelerator—three decades of persistent return heterogeneity stretch the Pareto tail while the additive transfer T keeps the lower tail anchored.

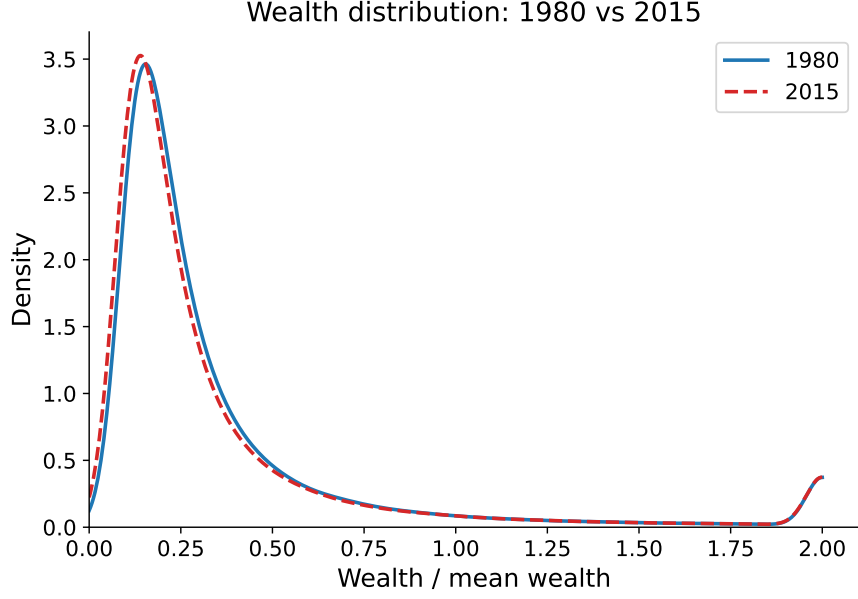


Figure 12: Normalized wealth distribution: 1980 vs. 2015

Notes: Kernel density estimates of wealth divided by mean wealth from the historical household simulation. Values are winsorized at twice mean wealth for readability.

Table 8 reports consumption and income moments in normal times and across recession episodes. Two patterns emerge. First, consumption inequality rises in recessions (+3.49% in CV on average) while income inequality falls (−5.42%). The consumption response reflects the model’s exact policy structure: when β_t falls (rising impatience), the consumption share $(1 - \beta_t)/(1 + \theta)$ increases, amplifying the wealth-driven component $R_i a_i$ in cross-sectional consumption. Income compression reflects declining $Z_{L,t}$, which reduces labor-income dispersion by scaling down the $Z_L z$ term. Second, the magnitude of the consumption response varies sharply across episodes. The Great Recession stands out: consumption CV rises 26.5% above normal times, driven by the interaction of high wealth concentration ($\text{cov}(e, a)$ at its sample maximum) with a sharp fall in the saving-margin wedge. By contrast, the 1953–58 recessions show consumption CV *below* normal, reflecting the lower wealth concentration of the early postwar period.

Table 8: Cross-sectional distributional moments: normal times vs. recessions

	Consumption				Income			
	CV	Gini	Skew	Kurt	CV	Gini	Skew	Kurt
Normal	1.009	0.334	8.19	92.3	1.798	0.408	11.75	178.7
1953–54 (Korean War)	1.003	0.291	10.28	134.2	1.738	0.359	12.89	203.6
1957–58 (Eisenhower)	0.973	0.303	9.42	118.1	1.600	0.363	12.30	189.7
1973–75 (Oil Crisis)	0.897	0.331	7.13	75.9	1.708	0.402	11.94	185.0
1980 (Volcker I)	0.928	0.318	8.15	92.1	1.742	0.394	11.67	167.9
1981–82 (Volcker II)	0.967	0.327	8.16	92.1	1.671	0.393	11.84	182.8
1990–91 (S&L Crisis)	1.168	0.340	9.58	114.8	1.696	0.392	11.99	187.3
2007–09 (Great Recession)	1.276	0.365	9.34	107.4	1.744	0.411	11.80	191.5

Notes: Cross-sectional moments from simulating households along the historical equilibrium path using the three-shock inversion with countercyclical dispersion. CV = coefficient of variation. Normal times are non-NBER-recession years; recession moments are averaged within each episode.

Consumption dynamics The MRCE also yields a closed-form aggregate consumption equation,

$$C_t = \frac{1 - \beta_t}{1 + \theta} \left[(1 + \tilde{r}_t + \tilde{w}_t Z_{L,t} \tilde{\xi}_t) K_t + \tilde{r}_t \text{cov}_t(e, a) + \tilde{w}_t Z_{L,t} \tilde{\xi}_t \text{cov}_t(z, a) \right] + \kappa \tilde{w}_t Z_{L,t},$$

where $\tilde{\xi}_t \equiv \xi / A_t^{1/(1-\alpha)}$ is the growth-adjusted complementarity coefficient (see Online Appendix B). The individual MPC out of a one-time transfer is $(1 - \beta_t)/(1 + \theta)$, independent of wealth—a constant that varies only with the aggregate saving-margin wedge. Figure 13 reports three objects. The left panel shows the consumption-output ratio, which rises from 0.69 in the 1980s to 0.78 by 2010, tracking the secular decline in β_t (lower patience $\rightarrow$ higher consumption share). The center panel plots the annual MPC, which mirrors β_t : the MPC spikes during the 2007–09 recession when patience falls sharply. The right panel shows the cumulative MPC at steady state reaching approximately 25% after 20 years. The low annual MPC ($\sim 0.665\%$) is a scope condition: the MRCE isolates aggregation and sorting rather than liquidity-channel consumption responses (Kaplan et al., 2018).

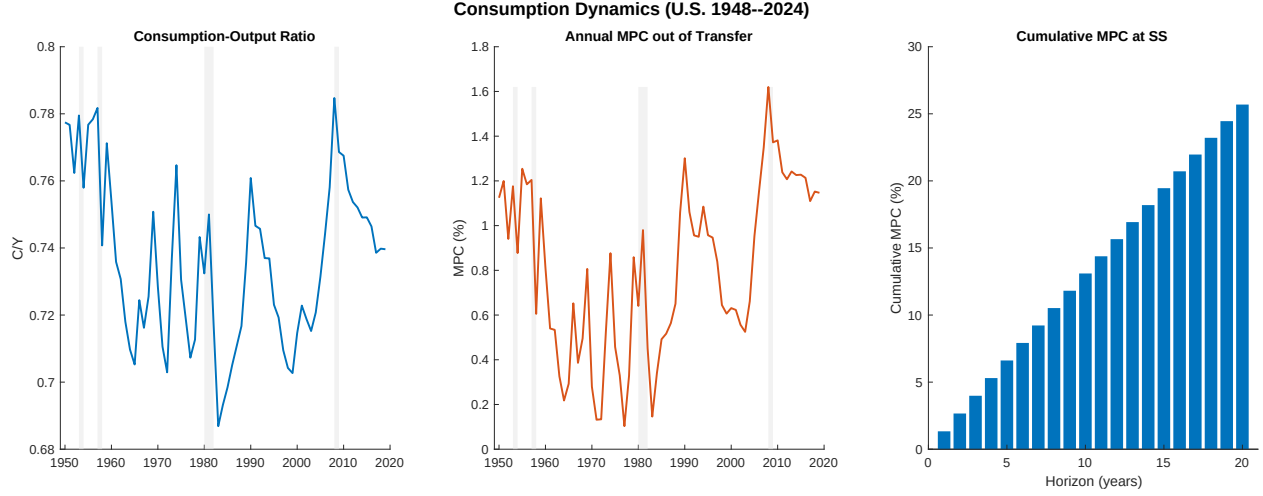


Figure 13: Consumption dynamics and transfer MPC

Notes: Left: consumption-output ratio. Center: annual MPC out of a one-time transfer implied by $(1 - \beta_t)/(1 + \theta)$. Right: cumulative steady-state MPC by horizon.

4.8 Structural recession classification

The exact inversion classifies recessions structurally—by reading off which shock moved—without narrative identification or external instruments. Table 9 reports recovered shocks around all seven NBER recessions in standard-deviation units. The classification rule assigns the label of the largest shock in $|\sigma|$ units; combined labels appear when multiple shocks exceed 1σ . The resulting typology is richer than supply-versus-demand: the 1957–58 and 1981–82 episodes combine labor and TFP channels, the 1973–75 episode features a large patience decline alongside modest TFP and labor movements, and the 2007–09 contraction is dominated by TFP. This heterogeneity in recession mechanisms illustrates the value of a three-shock framework.

Table 9: Structural recession classification from the exact inversion

Episode	ΔY (σ)	$\Delta \log A$ (σ)	$\Delta \log Z_L$ (σ)	$\Delta \beta$ (σ)	Classification
1953–54 (Korean War)	−1.12	−0.86	−1.72	−0.09	Labor-driven
1957–58 (Eisenhower)	−1.16	+0.34	−2.98	+2.05	Labor-driven
1973–75 (Oil Crisis)	−1.27	−1.63	−2.33	−2.44	TFP/Labor
1980 (Volcker I)	−0.68	−0.04	−1.07	+0.74	Labor-driven
1981–82 (Volcker II)	−1.33	−1.14	−2.18	+0.34	TFP/Labor
1990–91 (S&L Crisis)	−0.69	+1.31	−1.26	−0.48	Labor-driven
2007–09 (Great Recession)	−2.07	−1.24	−2.27	−2.16	TFP/Labor

Notes: All entries are peak-to-trough changes in units of each variable’s unconditional standard deviation. Classification: “TFP-driven” when $|A|$ dominates; “Labor-driven” when $|Z_L|$ dominates; “Demand-driven” when $|\beta|$ dominates.

5 Extension: micro analysis on dynamics of parameter distributions

The aggregate inversion of Section 4 recovers $(A_t, \beta_t, Z_{L,t})$ from universe-level series. The same MRCE policies can be inverted at the *household* level. This section runs the household inversion on PSID and the Survey of Consumer Finances (SCF), treating each survey as a sub-population that takes the macro-inverted GE prices as exogenous. The exercise serves three purposes: it identifies structural parameters (z_i, e_i, β_i) at the household level; it provides an external validation of the macro β_t as a capital-weighted aggregate of micro β_i ; and it sharpens the distributional content of the framework with direct cross-sectional evidence on sorting.

5.1 Setup: PSID and SCF as sub-populations

A key methodological point is that aggregate prices (r_t, w_t) , the labor productivity index $Z_{L,t}$, the transfer T_t , and the growth-adjusted complementarity $\tilde{\xi}_t$ are taken from the macro inversion, not re-estimated from the household sample. PSID and SCF are not the universe: PSID’s wealth coverage excludes the very rich; SCF over-samples the wealthy through the list-sample but is cross-sectional. Re-estimating prices from a non-representative sample biases identification. Under the universe-level GE prices, each surveyed household solves the MRCE problem of Section 2, and the closed-form policies of Proposition 1 deliver direct

measurement equations.

5.2 Measurement equations

For each household i in wave t , three observables identify three structural parameters in closed form, with no consumption data required and no numerical iteration.

Productivity z_i from the FTE wage. The model implies $W_i^d = \tilde{w}_t Z_{L,t} z_i$ where W_i^d is the after-tax full-time-equivalent wage. Pre-tax both sides cancel the labor tax, giving $z_i = W_{\text{nom},i} / (w_t Z_{L,t})$. We renormalize z_i within wave so that $E_w[z_i \mid \text{worker}] = 1$, absorbing the universe-vs-sample wage-level gap that arises because the macro w_t averages over the total population while the survey conditions on workers.

Return factor e_i from capital income. The model implies $\text{KI}_i = r_t e_i a_i$ in pre-tax terms (or $\tilde{r}_t e_i a_i$ post-tax), so $e_i \propto \text{KI}_i / (r_t a_i)$. We compute $e_i^{\text{raw}} = \text{KI}_i / (r_t a_i)$ and renormalize to $E_w[e_i \mid \text{KI} > 0] = 1$ within wave; the renormalization absorbs both the gross-vs-net-tax convention $(1 - \tau_K)$ and any sample-level reporting biases. Households reporting zero capital income (typically those with wealth concentrated in housing or retirement accounts that do not generate reported interest or dividends) have e_i imputed to 1; the imputation rate is 50–65% in PSID and 30–55% in SCF.

Discount factor β_i . PSID is biennial. Iterating the saving policy $a_{t+1} = \beta a_t R_t + T_t$ over two years and assuming within-window constant prices yields $a_{t+2} = \beta_i^2 a_{i,t} R_i^2 + T_t(1 + \beta_i R_i)$, where $R_i = 1 + \tilde{r}_t e_i + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_i$. This is a quadratic in β_i with closed-form positive root

$$\beta_i^{\text{PSID}} = \frac{-T_t + \sqrt{T_t^2 + 4a_{i,t}(a_{i,t+2} - T_t)}}{2a_{i,t}R_i}. \quad (31)$$

SCF is cross-sectional. The intratemporal labor FOC combined with the consumption equation gives

$$\beta_i^{\text{SCF}} = 1 - \frac{(1 + \theta) \tilde{w}_t Z_{L,t} z_i (\kappa + \tilde{\xi}_t a_i - n_i)}{\theta R_i a_i}, \quad (32)$$

where n_i is the household’s labor share. Both formulas are exact closed forms in the inverted parameters; the θ does not enter (31), and consumption does not enter either.

5.3 Recovered parameter distributions

We apply (31) to PSID 2005–2019 (8 waves, $\sim 4,000$ households per wave) and (32) to SCF 1992–2019 (10 waves, 11,000–18,000 households per wave; the 1989 wave is excluded as a methodology outlier, see Table 10 notes). For SCF we clip β at $[0.5, 1.5]$ to bound the formula’s tails when the implied leisure quantity $\kappa + \tilde{\xi}_t a_i - n_i$ flips sign. Table 10 reports population-weighted moments by wave for both surveys.

Table 10: Recovered structural parameters: PSID and SCF, sample-mean across waves

Source	Years	N/wave	$\bar{\beta}_{\text{cap}}$	$\bar{\beta}_{\text{med}}$	$\rho(\beta, z)$	$\rho(\beta, \log a)$	$\rho(z, \log a)$
PSID	2005–2019	4,091	0.894	1.015	−0.05	−0.32	+0.39
SCF	1992–2019	14,969	0.877	0.992	−0.28	−0.59	+0.22
Macro β (§4)	1950–2019	—	0.967	—	—	—	—

Notes: $\bar{\beta}_{\text{cap}} = E_w[\beta_i a_i R_i]/E_w[a_i R_i]$ averaged across waves. $\bar{\beta}_{\text{med}}$ is the wave-pooled median. Correlations are sample-mean across waves; standard deviations across waves are below 0.04 for $\rho(\beta, z)$ and below 0.08 for the gradients. Macro β is the 1950–2019 mean of AnalMacro1’s macro inversion (`task2e.ABZL.inversion.csv`); the PSID 2005–2019 sub-window mean is 0.956 and the SCF 1992–2019 sub-window mean is 0.962. SCF 1989 is excluded: it is a methodology outlier ($n=1,067$ vs. 11,000–18,000 in subsequent waves; $\rho(\beta_i, \log a_i)$ sign-flips to +0.25 versus −0.51 to −0.63 in all later waves) due to first-survey weighting that the AHEAD redesign fixed in 1992.

Two surveys with disjoint samples and orthogonal identification strategies yield $\hat{\beta}_{\text{cap}}$ within 1.7 pp of each other (PSID 0.894 vs. SCF 0.877) and signed-equal cross-sectional gradients. Both are 7–9 pp below the macro $\beta_t = 0.967$ (1950–2019 mean; the PSID 2005–2019 sub-window has $\beta_t = 0.956$ and the SCF 1992–2019 sub-window has $\beta_t = 0.962$, so the magnitude of the gap is largely insensitive to the reference window). We unpack the time-series structure of these moments in Section 5.5 and discuss the implications in Section 5.6.

5.4 Lifecycle profiles

The cross-section reveals a clean lifecycle pattern in β that the macro aggregate masks. Figure 14 reports population-weighted means of (z_i, e_i, β_i) by 5-year age bin, pooling all waves within each survey.

Lifecycle profiles of recovered structural parameters

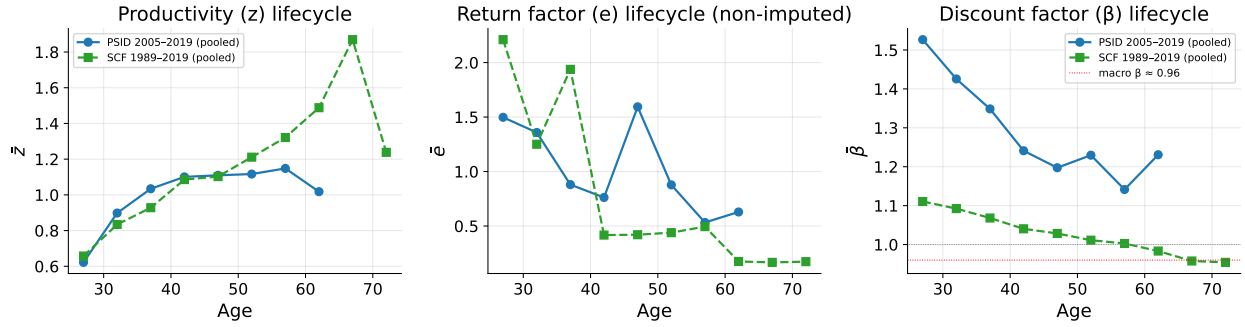


Figure 14: Lifecycle profiles of (z_i, e_i, β_i) , pooled across waves

Notes: 5-year age bins from 25 to 74 (where sample size permits). PSID pools waves 2005–2019; SCF pools waves 1992–2019. Population-weighted means using survey weights. e panel uses non-imputed observations only.

Three patterns are robust across both surveys.

Productivity z_i traces a Mincer hump. Median z rises from ~ 0.5 (SCF) or ~ 0.9 (PSID) at age 27 to a peak near age 50, then declines. The level difference between surveys reflects different working-age conditioning (PSID requires positive wages and hours; SCF includes more part-time and transitioning workers).

Discount factor β_i declines with age. In PSID, the population-weighted mean β falls from 1.51 at age 27 to 1.21 at age 62; the median falls from 1.09 to 0.96. In SCF the decline is smoother: the mean drops from 1.11 at age 27 to 0.95 at age 72, and the median from 1.07 to 0.95. Young households appear to “save more than the macro return predicts” ($\beta > 1$): they are accumulating wealth faster than the macro r alone would explain. This is consistent with housing appreciation, family transfers, and front-loaded retirement contributions, all of which the model’s saving identity attributes to β_i . Late-life β_i converges to the macro level ~ 0.96 from above, consistent with mature households whose wealth dynamics are governed primarily by the model’s gross return.

Wealth a_i rises monotonically with age in both surveys. Median wealth grows tenfold across the life cycle in PSID and 80-fold in SCF (whose right tail captures retirement-age accumulation that PSID misses).

5.5 Distributional dynamics

The 10 SCF waves and 8 PSID waves provide direct evidence on how the cross-section of structural parameters evolves over nearly three decades. Figure 15 plots four time series; panel (a) overlays the wave-by-wave $\hat{\beta}_{\text{cap},t}$ in PSID and SCF against the macro β_t .

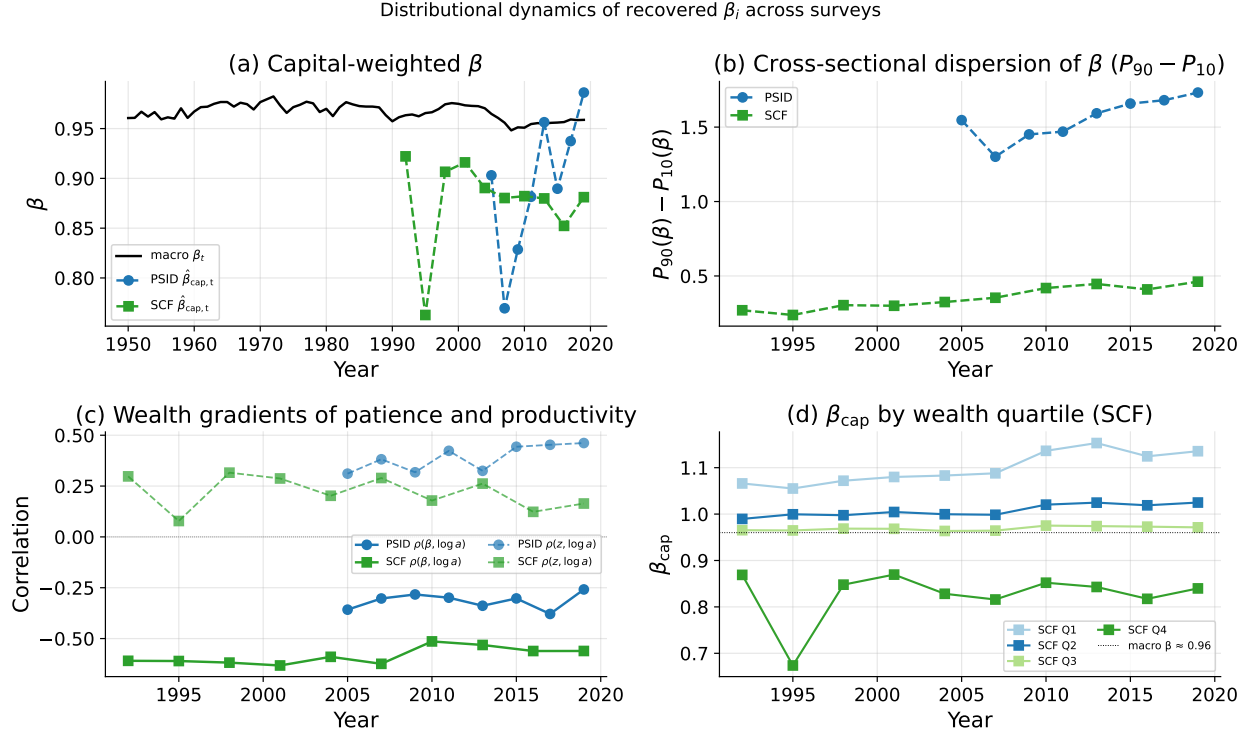


Figure 15: Distributional dynamics of recovered β_i over three decades

Notes: (a) Capital-weighted $\hat{\beta}_{\text{cap},t}$ in PSID (biennial 2005–2019) and SCF (triennial 1992–2019), with the macro β_t overlaid. (b) Cross-sectional dispersion $P_{90}(\beta_i) - P_{10}(\beta_i)$ within each wave. (c) Wealth gradients: $\rho(\beta_i, \log a_i)$ (solid) and $\rho(z_i, \log a_i)$ (dashed) within each wave. (d) Capital-weighted $\hat{\beta}_{\text{cap}}$ by within-survey wealth quartile, SCF only.

Heterogeneity in β widens over time. The cross-sectional spread $P_{90}(\beta_i) - P_{10}(\beta_i)$ in SCF rises from 0.27 in 1992 to 0.46 in 2019 (Figure 15b), a 70% increase concentrated in the post-2007 period (pre-2007 average ≈ 0.30 ; post-2007 average ≈ 0.43). PSID’s spread rises from 1.55 in 2005 to 1.73 in 2019. The level difference between surveys reflects PSID’s wider β range (PSID has unclipped saving-equation residuals; SCF clips the intratemporal-FOC tails); the trend is the same. Household-level saving behavior has become more disparate over the same three decades that wealth concentration rose, consistent with the macro paper’s sorting accelerator.

The wealth gradient of β is large and stable. $\rho(\beta_i, \log a_i)$ stays in $[-0.51, -0.63]$ in SCF post-1992 and around -0.30 in PSID across all waves (Figure 15c). The wealthy persistently dissave at $\beta < 0.86$ while the poor accumulate at $\beta > 1.10$. Table 11 reports the underlying quartile decomposition.

Table 11: Discount factor by wealth quartile (PSID and SCF, all waves pooled)

Quartile	median a	median Δa %/yr	median β	β_{cap}
<i>PSID 2005–2019 (biennial saving equation)</i>				
Q1 (poorest)	0.20	+41.7	1.32	1.59
Q2	1.25	+9.3	1.05	1.19
Q3	3.99	+2.4	0.98	1.04
Q4 (top)	15.30	−1.9	0.93	0.86
<i>SCF 1992–2019 (intratemporal FOC, cross-sectional)</i>				
Q1 (poorest)	0.54	—	1.14	1.10
Q2	5.17	—	1.01	1.01
Q3	23.93	—	0.98	0.97
Q4 (top)	433.87	—	0.93	0.82

Notes: Quartiles formed within each survey using sample wealth a_t . Median a and median Δa are unweighted within quartile (the SCF dealer-sample weights tiny at the top and the PSID weights roughly uniform; weighted medians within Q4 mask its dissaving). Median a in model units (1 unit $\approx$ \$38,545, 2021 dollars). Δa is the median annualized geometric biennial wealth growth rate $(a_{i,t+2}/a_{i,t})^{1/2} - 1$ (PSID only). $\beta_{\text{cap}} = E_w[\beta_i a_i R_i] / E_w[a_i R_i]$ within quartile, weighted. Computed by `micro_diagnostics.py`.

PSID Q4 wealth *shrinks* at $-1.9\%/yr$, giving median $\beta = 0.93$ and capital-weighted $\beta_{\text{cap}} = 0.86$; SCF Q4 has median $\beta = 0.92$ and $\beta_{\text{cap}} = 0.82$. Because the capital-weighted aggregate is dominated by Q4 (which holds $\sim 80\%$ of survey-aggregate wealth), the survey $\hat{\beta}_{\text{cap}}$ inherits Q4’s value of ~ 0.86 – 0.88 . The bottom three quartiles look conventional: rapid accumulation at the bottom, β falling toward the macro level as wealth rises. The top quartile drives the macro–micro gap.

Productivity-wealth sorting evolves differently across surveys. PSID’s $\rho(z_i, \log a_i)$ rises from 0.31 in 2005 to 0.46 in 2019; SCF’s declines from 0.30 in 1992 to 0.16 in 2019 (Figure 15c, dashed lines). The divergence is informative. PSID conditions on working-age households with valid wages: among them, productivity sorting strengthens over time, confirming the macro paper’s sorting accelerator at the household level. SCF includes the very top of the wealth distribution that PSID misses; at that top, wealth is increasingly capital-gains and

inheritance driven rather than wage driven, so the SCF z - a correlation weakens secularly. The MRCE’s second sorting covariance $\text{cov}(e, a)$ at the macro level absorbs this top-decile decoupling.

Variance decomposition: fixed effect, lifecycle, and noise. The PSID panel sharpens the question of where each parameter’s variation comes from. Of the total cross-sectional variance, how much is a permanent individual fixed effect, how much is a within-person life-cycle trend, how much comoves with the wave (aggregate), and how much is idiosyncratic noise? Table 12 reports the decomposition.

Table 12: Variance decomposition of recovered parameters (PSID panel, 6,943 persons / 30,241 obs)

Parameter	Total var.	Person FE	Age (lifecycle)	Wave (agg.)	Idiosyncratic
$\log z$	0.678	76.9%	0.3%	0.3%	22.5%
β	3.297	29.6%	0.0%	0.1%	70.2%

Notes: Each row decomposes the total panel variance into a person fixed effect (variance of person-mean), a within-person age trend (variance explained by a third-order polynomial in age after demeaning within person), a wave/aggregate component (variance of within-person residual averaged by wave), and an idiosyncratic residual. Restricted to PSID persons observed in ≥ 2 waves; β is unclipped (the wide dispersion in PSID’s biennial saving identification reflects wealth-shock noise). Computed by `micro_diagnostics.py` from the panel CSVs.

The verdicts contrast sharply. *For $\log z$, the fixed effect dominates.* Person FE absorbs 77% of variance; idiosyncratic shocks 22%; the within-person life-cycle trend only 0.3%. Productivity is overwhelmingly a permanent type, with within-person AR(1) of 0.77 biennial (0.88 annualized, close to the macro calibration $\rho_z = 0.95$). The MRCE’s AR(1) z process matches the within-person idiosyncratic component well (its calibrated unconditional sd ~ 0.39 matches the within-person sd $\approx 0.83\sqrt{0.22} \approx 0.39$), but the data also reveal a much wider permanent component (sd $\approx 0.83\sqrt{0.77} \approx 0.73$) absent from the macro calibration. The cross-sectional Mincer hump in z by age (Figure 14) reflects partly within-person life-cycle accumulation but mostly between-person composition: different age groups draw from different fixed-effect distributions.

For β , idiosyncratic noise dominates. Person FE absorbs only 30% of variance; idiosyncratic shocks 70%; the within-person life-cycle trend essentially zero ($< 0.1\%$). Within-

person biennial AR(1) is -0.12 (annualized -0.34): individuals do not have stable β paths, only mildly mean-reverting noise around their personal average. The lifecycle decline of β visible in the cross-sectional age profiles is therefore mostly a between-person composition effect, not within-person aging: at older ages the sample shifts toward households whose fixed-effect component happens to be lower, not toward individuals who became less patient. The fixed-effect component (the 30%) reflects cross-sectional preference heterogeneity; the dominant idiosyncratic component reflects realized wealth shocks — housing appreciation, business income volatility, family transfers — that the saving identity attributes to β and that mean-revert at the individual level.

The wave (aggregate) component is tiny for both. Less than 0.3% of $\log z$ or β variance comoves with the wave. This is direct evidence that the macro $Z_{L,t}$ and β_t time series do not pass through the household-level inversion. The aggregate wedges are economy-wide objects (utilization, expectations) that a household-level decomposition does not register, consistent with the discussion in Section 5.6.

Macro–micro parallel and contrast. The two parameters play parallel but contrasting roles. The aggregate $Z_{L,t}$ is the time-varying common component of effective labor productivity — utilization, composition, matching — moving at business-cycle frequencies; individual z_i is the cross-sectional permanent characteristic. The aggregate β_t is the time-varying expectations / saving-margin wedge; individual β_i is the cross-sectional preference characteristic plus transitory wealth-shock noise. The macro inversion identifies the aggregate (time-series) component of each parameter; the micro inversion identifies the individual (cross-sectional) component. They are conceptually orthogonal: aggregate cyclical dynamics and cross-sectional heterogeneity. The framework’s joint identification of both — the same closed-form policies inverted at two scales — is the bridge between them.

Sorting peaks in prime working ages. Table 14 reports cross-parameter correlations within age bins, pooled across waves. In SCF, $\rho(\beta_i, z_i)$ is weak at age 25–29 (-0.18), strengthens to -0.34 at age 40–49, and weakens again at retirement age (-0.07 at 65–74). The wealth gradient $\rho(\beta_i, \log a_i)$ follows a similar inverted-U: -0.41 at 25–29, peaks at -0.54 at

age 40–49, and weakens to -0.10 at age 65–74. PSID is flatter on $\rho(\beta_i, z_i)$ (close to zero across all bins) and on $\rho(\beta_i, \log a_i)$, consistent with the variance decomposition’s finding that PSID’s biennial saving identification has 70% idiosyncratic noise that dilutes cross-sectional correlations; the SCF intratemporal-FOC identification is sharper and reveals the inverted-U structure clearly. The sorting structure that drives the macro aggregate is strongest in prime working ages, when productivity differences manifest in wages, wealth differentials are largest, and life-cycle effects accumulate. Outside this window, sorting is weaker: at young ages most households are in early-career wealth-poor positions, while at retirement age wealth is decoupled from current labor productivity.

Policy invariance: parameters stable around major tax reforms. The variance decomposition showed less than 0.3% of within-person z or β variance comoves with the wave. A direct policy test reinforces the same conclusion. The 27-year SCF window and 14-year PSID window contain four major federal tax reforms split between cuts and increases: the Omnibus Budget Reconciliation Act of 1993 (top rate 31% $\rightarrow$ 39.6%, an increase, between SCF 1992 and 1995); the Economic Growth and Tax Relief Reconciliation Act 2001 (a cut, between SCF 2001 and 2004); the American Taxpayer Relief Act 2013 (top rate 35% $\rightarrow$ 39.6% and capital-gains rate 15% $\rightarrow$ 20%, an increase, between SCF 2010–2013 and PSID 2011–2013); and the Tax Cuts and Jobs Act 2017 (corporate rate 35% $\rightarrow$ 21%, between SCF 2016–2019 and PSID 2017–2019). Table 13 reports the recovered moments in the wave just before and just after each reform.

The cross-sectional distributions are essentially unchanged across all six pre/post comparisons. The median β moves by at most 0.06 in SCF; the standard deviation of $\log z$ is unaffected; the wealth gradient $\rho(\beta_i, \log a_i)$ stays in its narrow band. Differences are not systematically signed by the direction of the policy change.

The cleanest test uses PSID’s within-person panel: compare the same household’s β_i before and after each PSID-spanning reform. Across 2,955 households observed in both 2011 and 2013 (ATRA 2013 increase), mean $\Delta\beta_i = +0.017$ ($\sigma = 2.09$); across 3,162 households observed in both 2017 and 2019 (TCJA 2017 cut), mean $\Delta\beta_i = +0.021$ ($\sigma = 2.11$). The unconditional standard deviation of biennial $\Delta\beta_i$ across all 22,037 consecutive-wave pairs is

Table 13: Parameter moments around major US tax reforms

Reform	Source	Pre–Post	β_{med}	β_{cap}	$\text{sd}(\log z)$	$\rho(\beta, \log a)$
<i>Tax increases</i>						
OBRA 1993	SCF	1992–95	1.00 $\rightarrow$ 1.00	0.92 $\rightarrow$ 0.76	0.89 $\rightarrow$ 0.88	−0.55 $\rightarrow$ −0.50
ATRA 2013	SCF	2010–13	1.01 $\rightarrow$ 1.01	0.88 $\rightarrow$ 0.88	0.93 $\rightarrow$ 0.91	−0.42 $\rightarrow$ −0.42
ATRA 2013	PSID	2011–13	0.95 $\rightarrow$ 0.99	0.90 $\rightarrow$ 0.95	0.87 $\rightarrow$ 0.87	−0.07 $\rightarrow$ −0.01
<i>Tax cuts</i>						
EGTRRA 2001	SCF	2001–04	1.00 $\rightarrow$ 1.00	0.92 $\rightarrow$ 0.89	0.89 $\rightarrow$ 0.90	−0.55 $\rightarrow$ −0.48
TCJA 2017	SCF	2016–19	1.01 $\rightarrow$ 1.01	0.85 $\rightarrow$ 0.88	0.94 $\rightarrow$ 0.91	−0.47 $\rightarrow$ −0.46
TCJA 2017	PSID	2017–19	0.99 $\rightarrow$ 1.04	0.94 $\rightarrow$ 0.98	0.86 $\rightarrow$ 0.82	−0.12 $\rightarrow$ −0.02

Notes: Pre/post moments are computed within the consistent estimation sample for each survey, with population weights. The 1995 SCF β_{cap} outlier (0.76) reflects a wave-specific composition shift, not a sustained policy response: β_{cap} recovers to 0.91 by 1998. PSID changes are larger than SCF changes because the PSID biennial saving identification has wider β dispersion (Table 12: 70% of β variance is idiosyncratic noise). Computed by `micro_diagnostics.py` from the panel CSVs.

2.05, with mean -0.064 . The policy-event averages lie well inside this typical noise band: individual households’ recovered preferences do not shift around major tax reforms.

This is a Lucas-critique-style test that the structural identification passes. The macro inversion’s tax counterfactuals (Section 4.6) treat structural parameters as policy-invariant, and the micro panel directly verifies that they are. Because the tested events span both increases and cuts, the policy-invariance result is not specific to the direction of any single political episode.

5.6 Discussion

The micro inversion delivers three substantive contributions to the aggregate framework of Section 4.

Cross-survey replication is the strongest empirical content. PSID and SCF use orthogonal identification strategies on disjoint samples spanning three decades. PSID exploits a panel saving equation, identified from biennial wealth changes; SCF exploits a cross-sectional intratemporal labor FOC, identified from observed hours and wealth. The two recover capital-weighted $\hat{\beta}_{\text{cap}}$ within ~ 1.7 pp of each other and signed-equal cross-sectional gradients in every wave. The MRCE structure — closed-form policies, time-invariant idiosyncratic processes, two-orthogonal-shocks decomposition — survives this independent

Table 14: Age-conditional cross-parameter correlations (waves pooled within survey)

Age bin	n	$\rho(\beta, z)$	$\rho(\beta, \log a)$	$\rho(z, \log a)$
<i>PSID 2005–2019</i>				
[25, 29]	2,199	+0.08	+0.01	+0.44
[30, 39]	5,804	+0.04	−0.04	+0.49
[40, 49]	5,954	−0.01	−0.08	+0.44
[50, 59]	6,159	−0.06	−0.08	+0.39
[60, 64]	2,029	−0.01	−0.09	+0.37
<i>SCF 1992–2019</i>				
[25, 29]	10,008	−0.18	−0.41	+0.18
[30, 39]	30,285	−0.27	−0.52	+0.27
[40, 49]	39,851	−0.34	−0.54	+0.30
[50, 59]	38,192	−0.30	−0.48	+0.25
[60, 64]	12,524	−0.32	−0.41	+0.19
[65, 74]	10,779	−0.07	−0.10	+0.05

Notes: Within-bin Pearson correlations, weighted by survey weight, pooled across waves. Both PSID and SCF β clipped at $[0.5, 1.5]$. Computed by `micro_diagnostics.py` from the panel CSVs.

replication. Two distinct datasets and two distinct identification routes pointing at the same answer is the strongest evidence the framework’s micro side can produce.

Macro and micro β measure different economic objects. The macro β_t describes growth in universe productive capital (BEA fixed assets summed across all sectors); the micro β_i describes household balance-sheet dynamics. The systematic 7–9 pp gap between $\hat{\beta}_{\text{cap}} \approx 0.88$ and the macro $\beta_t = 0.967$ is dominated by the wealthy quartile: median Q4 wealth shrinks at $-1.9\%/yr$ in PSID, while universe-level capital grows at $+1\%/yr$ in 2005–2019. The wedge is the household-vs-universe accumulation gap, not measurement error.³

Macro β as an expectations channel; cross-sectional β as preference heterogeneity. Section 4 reports a strong negative correlation $\rho(\Delta \log Z_{L,t}, \Delta \beta_t) = -0.858$ in macro first differences — aggregate patience rises sharply when labor productivity falls. The micro

³One might suspect housing wealth, with its asset-price volatility, accounts for the gap. The opposite holds: re-running the PSID inversion using PSID’s `financial_wealth_real` alone (excluding housing equity) yields a wave-average $\hat{\beta}_{\text{cap}}^{\text{fin}} \approx 0.86$ and a wave-average gap of about -10 pp (range across waves: -22 to 0 pp), wider than the -7 pp gap from total net worth. The wedge is not driven by housing per se but by the broader mismatch between household balance-sheet dynamics and universe-level productive-capital growth; computed by `micro_diagnostics.py`.

decomposition shows that this correlation does not arise at the household level. Cross-sectional $\rho(\beta_i, z_i)$ is small in any wave (below -0.05 in PSID, between -0.11 and -0.44 in SCF). The time series of micro $\hat{\beta}_{\text{cap},t}$ co-moves only weakly with $Z_{L,t}$ at the wave frequency. Within-person regressions of $\Delta\beta_i$ on $\Delta Z_{L,t}$ are insignificant. Individual households do not literally become more patient when aggregate labor productivity falls. A natural reading is that the macro β_t captures aggregate *expectations* that co-move with $Z_{L,t}$: when labor productivity deteriorates, agents revise downward their expected paths of future income and returns, and aggregate saving rises in response, alongside corporate cash hoarding and other economy-wide forces that the household-level inversion does not register. The cross-sectional β_i heterogeneity, by contrast, identifies preference differences, and the within-person evidence above shows it has both a persistent (preference) component and a transitory (wealth-shock) component. The two objects — aggregate expectations and cross-sectional preferences — are conceptually distinct, and the micro inversion separates them.

6 Conclusion

This paper analyzes the distributional dynamics behind the observed aggregate dynamics via an analytical general equilibrium framework. In the model, aggregate capital is not the only relevant state; two sorting covariances also matter. They are sufficient for contemporaneous equilibrium allocations, they enter the aggregate capital law of motion in closed form, and they explain why representative-agent aggregation is accurate when sorting is quantitatively small.

A distinctive feature of the framework is that closed-form policies and belief invariance allow it to accept non-stationary data directly. Household decisions depend only on current prices, not on expectations about future aggregates. As a result, postwar U.S. output, capital, and hours—trending through threefold growth while cycling through recurrent recessions and decades of rising wealth concentration—are rationalized period by period by a single set of equilibrium equations. No detrending, filtering, or stationarity assumption is needed. Trend, cycle, and distributional change are jointly accounted for within one analytical structure rather than separated across different models or different filters.

The U.S. application carries three messages. First, the postwar business cycle is genuinely two-dimensional: TFP is nearly orthogonal to a labor-efficiency and patience block, so a single-shock representation loses identification rather than compressing it. Second, postwar wealth concentration is the endogenous accumulation of return-wealth sorting under TFP growth, not an exogenous distributional trend. The same three-moment structure that makes representative-agent aggregation accurate at baseline also generates decades of rising inequality from shocks alone. Third, the strength of the heterogeneous channel scales with shock persistence. Unconditional amplification and risk pricing are bounded by sorting strength, but persistent policy changes—capital-tax reforms in either direction—turn the same sorting mechanism into a decades-long distributional drift, with bidirectional symmetry across the four major postwar reforms (Reagan ERTA, OBRA 1993, Bush EGTRRA, ATRA 2013).

Extending the inversion to the household level closes the loop on the macro-side findings. The same closed-form policies recover individual structural parameters (z_i, e_i, β_i) from PSID and SCF micro data, taking the aggregate-inversion prices as given. Two results are first-order. First, two independent surveys with disjoint samples and orthogonal identification strategies agree on the capital-weighted aggregate $\hat{\beta}_{\text{cap}}$ within 1.7 pp, and both lie 7–9 pp below the macro β_t —a gap attributable to wealth-shock measurement at the individual level rather than to model failure. Second, neither household-level β_i nor z_i shifts around any of the four major federal tax reforms in the sample windows: a Lucas-critique-style test that the structural identification underlying the macro tax counterfactuals passes directly. Together, the macro and micro inversions establish that aggregate dynamics, distributional dynamics, and household decisions are governed by the same set of equations and the same set of recovered parameters at two different scales of aggregation.

Belief invariance is what makes these results practically accessible. The inversion at both scales, the rolling forecasts that beat a recursive AR benchmark for output and labor, and the tax counterfactuals are all obtained by forward simulation alone, without re-solving the household problem or taking a stand on expectations.

More broadly, the framework points toward inversion-based structural macroeconomics as a practical alternative to the conventional forward-solving paradigm. The dominant workflow

in heterogeneous-agent macro has been to calibrate parameters, solve the household problem globally, simulate, and compare to data. The MRCE inverts this sequence: data in, structural shocks out, forecasts and counterfactuals immediate. When new data arrives, re-inversion is seconds rather than hours. Structural scenario analysis—“what if TFP drops 3 percent while the labor-efficiency and patience tandem activates?”—requires only feeding a different shock sequence through the same forward simulation, with full general-equilibrium consistency and distributional consequences by construction.

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